
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

Anonymous Authors

ABSTRACT

Uniform manifold approximation and projection (UMAP) is among the most popular neighbor embedding methods. The method relies on attractive and repulsive forces among high-dimensional data points to obtain a low-dimensional embedding. In this paper, we analyze these forces to quantify their effects on cluster formation and visualization, benchmarking UMAP directly against its contemporaries (NEG-*t*-SNE, PaCMAP, and others) under matched conditions. Repulsion emphasizes differences, controlling cluster boundaries and inter-cluster distance. Attraction is more subtle: attractive tension between points can manifest simultaneously as attraction and repulsion in the lower-dimensional mapping. This dual behavior precisely explains why learning rate annealing is needed and motivates the different treatments between attractive and repulsive terms. Building on this diagnosis, a single modification to the attraction shape yields a substantial and consistent gain: across three datasets and 100 random initializations each, it reduces the mean Procrustes distance to a PCA-initialized reference by roughly one-third or more (e.g., from 0.78 to 0.49 on MNIST), sharply improving the run-to-run consistency of cluster formation. Together, these measured comparisons make UMAP and similar embedding methods more interpretable, more robust, and more accurate.

1 Introduction

The current era is characterized by a deluge of high-dimensional data. Dimensionality reduction (DR) techniques have emerged as tools for exploratory analysis of such data by visualizing the underlying structure. The most popular methods, *t*-distributed stochastic neighbor embedding (Maaten and Hinton [2008]) and uniform manifold approximation and projection (UMAP) (McInnes et al. [2018]) are grounded in the attraction-repulsion dynamics that bring similar data points closer while pushing dissimilar ones apart. As unsupervised algorithms, these do not rely on labeled data; instead, they identify and preserve the intrinsic structure of high-dimensional data by leveraging local (attractive) and global (repulsive) relationships (forces). This makes these algorithms particularly well-suited for tasks such as clustering (Becht et al. [2019]), exploratory data analysis (Fleischer and Islam [2020]), anomaly detection in semiconductor manufacturing (Fan et al. [2021]), visual search (González-Márquez et al. [2024]), time series analysis (Altin and Cakir [2024]), studying representation convergence (Huh et al. [2024]), and outlier image detection (Islam and Fleischer [2024]), where visualizing hidden patterns in unlabeled data is critical and meaningful. By learning the embeddings in a data-driven, label-free manner, DR exemplifies the power of unsupervised methods to distill complex data into easily interpretable forms.

Building upon the attraction-repulsion principle, newer methods have emerged (Amid and Warmuth [2019], Agrawal et al. [2021], Wang et al. [2021], Narayan et al. [2021], Yang et al. [2022], Wang et al. [2025], Kury et al. [2025]), each designed to emphasize specific aspects of the data. Despite their relevance in diverse applications, these methods often rely on heuristic practices that may fail to give meaningful interpretations. Moreover, DR introduces distortions that are unavoidable (Chari and Pachter [2023]). A deeper, quantitative understanding of the algorithms is therefore imperative, so that practitioners can better interpret embeddings, avoid spurious structures, and optimize performance on measurable criteria. In practice, these algorithms achieve compact clusters using a variety of techniques, including specific initialization, learning rate schedule, and kernel function tuning. Yet the underlying dynamics of the attractive and repulsive forces responsible for cluster formation have not been thoroughly investigated, and the essential tunable parameters remain concealed within abstract functional forms. This gap makes it difficult to say precisely which force drives which measured outcome, or by how much one method improves over another.

In this paper, we decompose the forces into their constituent parts and extract the functional shapes of attraction and repulsion. This decomposition lets us attribute each observed empirical behavior to a specific force: we find that the necessity of learning rate annealing, the challenge of providing consistent output under random initialization, and the origin of cluster formation all trace back to attraction, whereas repulsive forces primarily govern inter-cluster distances. Crucially, these attributions are not qualitative claims alone; each is backed by controlled comparisons in which we hold one force fixed and vary the other, and by quantitative metrics (trustworthiness, silhouette score, and Procrustes distance) measured across UMAP, NEG- t -SNE, and PaCMAP on multiple datasets. Our specific contributions are:

1. We formulate attraction and repulsion shapes from the attractive and repulsive forces, establish the conditions for contraction and expansion of distance, and provide a fresh perspective for these algorithms (Section 4).
2. We show that the attraction shape of UMAP causes the counterintuitive concept of both contraction and expansion of distance. Comparing attraction shapes of different algorithms, we discuss how attraction influences the learning rate annealing scheme (Section 5).
3. We modify the attraction shape and measure a clear, consistent improvement in embedding consistency under random initialization—the mean Procrustes distance to a PCA-initialized reference drops by roughly one-third or more on every dataset tested—indicating the encoding of a unique structure (Section 5.1).
4. Analyzing repulsion shapes, we provide a deeper understanding of cluster formation and regulating inter-cluster distance (Section 5.2).
5. We place the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE side by side, quantifying which shapes stay within the stability-guaranteeing range $[-1, 0]$ and which do not, and thereby characterize the relative stability of the algorithms under a fixed learning rate (Section 6).

We center the main text on UMAP, now a de facto standard in many fields, and bring in other algorithms where relevant. For further details on different algorithms, see the appendix. The codes used in the research will be released upon acceptance.

2 Related Works

The origin of modern iterative graph-based neighbor embedding algorithms can be traced to stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its extension using the t -distribution (t -SNE) (Maaten and Hinton 2008). Both methods use a dense graph in which each point in a dataset has a pairwise relation with all the others, regardless of whether they are similar to each other or not. Moreover, the weights of the graphs are normalized to give a notion of probability distribution. Other concurrent methods, including locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002), used a k -nearest neighbor (k -NN) graph of pairwise interaction. Known as spectral methods, these algorithms rely on Eigenvalue decomposition. Subsequent work by Tang et al. (Tang et al. 2016) incorporated the k -NN graph in the iterative approach and removed normalization in the lower dimension. This approach was further extended by McInnes et al. (2018) in UMAP, where the normalization step was removed altogether (both in high and low dimension) and the embedding was obtained using pairwise interactions alone. The optimization steps use an explicit attractive force to preserve the local neighborhood and a repulsive force to keep dissimilar points apart. Building on these foundations, methods such as PaCMAP (Wang et al. 2021) and NEG- t -SNE (Damrich et al. 2023) have been proposed. For a recent survey of methods, see de Bodt et al. (2025).

There has been considerable progress in understanding and explaining the relationship among these algorithms, much of it framed as quantitative guarantees. An early analysis of SNE established that if the data is well-clustered in the original space, then it remains well-clustered in the embedding space (Shaham and Steinerberger 2017). A parallel analysis for t -SNE by Linderman and Steinerberger (2019) sharpened this into an explicit bound: the number of clusters in the embedding space is a lower bound on the number of clusters in the original space. This line was followed up with further quantitative characterizations (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Since t -SNE and UMAP originate from the same underlying framework (but with drastically different visualizations), a major undertaking in the literature has been to find the connection between them (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) theorized that methods like Laplacian Eigenmap, ForceAtlas2 (Jacomy et al. 2014), UMAP, and t -SNE are all samples from the same underlying spectrum. Indeed, LE and t -SNE’s are connected by the early exaggeration phase (Cai and Ma 2022). The connection between UMAP and t -SNE can be related through contrastive estimation (Damrich et al. 2023). Around the same time, Hu et al. (2023) independently discovered the relation of contrastive learning and SNE. Recent approaches offer a probabilistic perspective (Ravuri et al. 2023; Ravuri and Lawrence 2024), employ kernel techniques (Draganov and Dohn 2023), and utilize information geometry (Kolpakov and Rocke 2024) to explain dimensionality reduction.

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by using a pairwise relation: $p_{i,j} = f_h(d(x_i, x_j))$, typically, $\in [0, 1]$ (for details, see Appendix H).

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is given by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points.

UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term provides an attractive force and the second term provides a repulsive force. Rather than optimizing every point in each iteration, UMAP takes the negative sampling approach (Mikolov et al., 2013; Tang et al., 2016): for each edge with $p_{ij} > 0$, named a positive edge, several edges are sampled randomly, named negative edges. This asymmetric treatment of the two terms is exactly the design choice our analysis later explains quantitatively. The attractive force is applied on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force is applied on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. Note that y_j is not updated for negative edges. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The action of the updates (4, 6) can be simplified by decomposing the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) into a scalar coefficient dependent on the distance $\|y_i - y_j\|_2$ acting on the vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) shape. While we use UMAP as a specific example, this formalism applies to any method that relies on attraction and repulsion.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we use the fact that for Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. Similarly, we reformulate the update equation of a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Such decomposition has appeared previously, e.g., in (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023), but their formulations and utilization differ from ours in a way that matters for what can be measured. The original UMAP paper (McInnes et al., 2018) used it as a computational trick for fast processing, while Draganov and Dohn (2023) used it to compare different algorithms from the kernel perspective; in both cases the primary focus was computing the derivative. Here, by contrast, we treat the decompositions as independent functions that can take various forms. The closest prior approach, by Agrawal et al. (2021), expressed the decomposition as a scalar coefficient times a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize the magnitude of the forces alone. The comparisons below show concretely that our shape decomposition is more illuminating than magnitude alone, because the sign and range of the shape—not just its size—predict whether a pair contracts or expands.

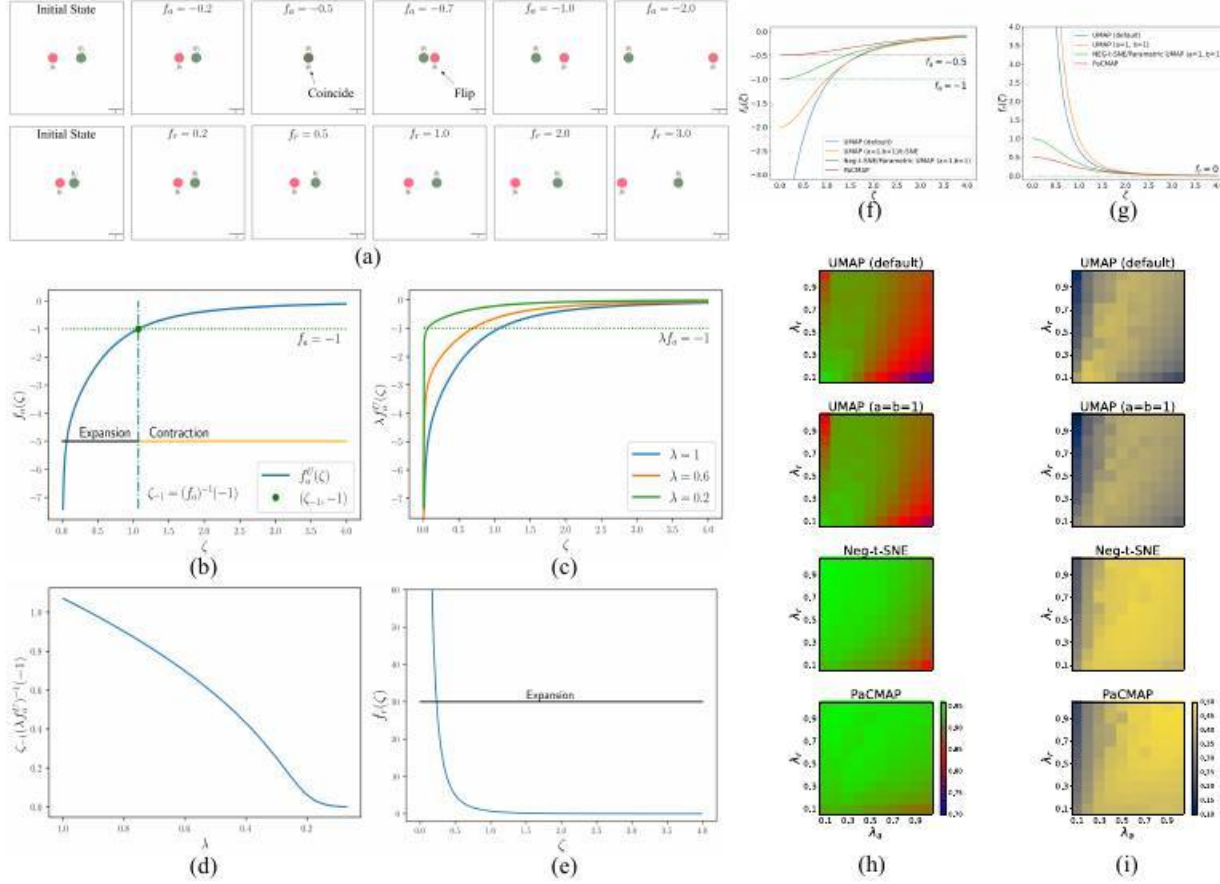


Figure 1: Attraction and repulsion shapes in UMAP, and a head-to-head comparison across methods. (a) Effect of different values of f_a (top) and f_r (bottom) on a pair. (b) Attraction shape of UMAP. (c) Effective attraction shape (λf_a) for various learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes overlaid for all methods, making explicit that only UMAP and t -SNE fall below -1 . (h) Trustworthiness and (i) Silhouette Score for each method as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r): the two metrics jointly show that UMAP degrades once $\lambda_a \gtrsim 0.5$, whereas NEG- t -SNE and PaCMAP remain stable across the full range. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

In this section, we establish the conditions of attraction and repulsion from the update equations (7)–(10). The following proposition characterizes the contraction of distance between the pair y_i^t and y_j^t of a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here, λf_a works as the effective attraction shape. In most cases, however, f_a alone is enough to draw meaningful conclusions about the embeddings. For a negative edge, the following proposition characterizes the expansion of the distance between the pair y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Note that the inclusion of a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) does not alter the conclusion presented in this proposition. Importantly, these two conditions give an exact, per-iteration certificate for guaranteed contraction or expansion—a sharper statement than a learning-rate heuristic. They should not be conflated with the learning-rate tuning/decay mechanism: one can design shapes for which the contraction and expansion hold regardless of learning-rate decay, a distinction we exploit quantitatively when comparing methods below.

Figure 1(a) contrasts the effect of different values of $f_a < 0$ and $f_r > 0$ on two points. The repulsion case is straightforward: any positive value increases the distance. The attraction case is far more subtle, because f_a encodes both attractive and repulsive dynamics within a single sign-restricted range. Concretely, only the narrow window $f_a \in (-1.0, 0)$ contracts the pair, with the point of maximum attraction (coincident points) at exactly $f_a = -0.5$; the interval crosses over so that any value below -1.0 instead expands the distance. This exact threshold structure—not the magnitude of f_a —is what governs behavior, and although the forces act locally, they collectively shape the global structure.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form 1, the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section focuses on the default shapes of UMAP, controlled primarily by the parameters a and b , and discusses the insights learned by perturbing them. The discussion below is generally valid for $b \leq 1$, where the attraction shape is strictly increasing and thus invertible (For derivation, see Appendix A).

Figure 1(b) shows the default attraction shape of UMAP ($a = 1.58, b = 0.89$), which becomes unbounded (approaches ∞) as $\zeta \rightarrow 0$. As predicted quantitatively by Proposition 4.1 the transition from contraction to expansion occurs exactly when λf_a^U crosses -1 as ζ approaches 0. Because the attraction shape is invertible, we can pin this transition to a specific distance, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, the minimum distance for contraction due to attractive updates. The prediction is sharp: $\zeta > \zeta_{-1}$ contracts, while $0 < \zeta < \zeta_{-1}$ expands—a directly measurable reversal that contradicts the intuition that attractive updates consistently bring points closer together.

If ζ_{-1} is high, neighboring points oscillate between contraction and expansion, and the clusters appear fuzzy. For the sharpest boundaries, then, the goal of optimization can be recast as one of achieving the limit $\zeta_{-1} \rightarrow 0$. As a result, UMAP’s learning rate schedule requires annealing to zero (Figs. 1(c,d)). On the other hand, the repulsion shape (Eq. 12) is always positive and satisfies Proposition 4.2 f_r^U approaches 0 as $\zeta \rightarrow \infty$ and approaches ∞ as $\zeta \rightarrow 0$ (Fig. 1(e)).

Overlaying the attraction shapes of all algorithms (Fig. 1(f)) makes the comparison unambiguous: only UMAP and t -SNE cross below -1 (and consequently have $\zeta_1 > 0$ at $\lambda = 1$), whereas NEG- t -SNE and PaCMAP remain entirely within $[-1, 0]$. The two methods that cross below -1 compensate in different ways— t -SNE by weighting updates with the corresponding p_{ij} values, UMAP by learning-rate annealing—while NEG- t -SNE and PaCMAP satisfy Proposition 4.1 for the whole range $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$ and thus need no such correction. PaCMAP goes further still: its weighted f_a stays below 0.5 in magnitude at all times, ruling out flips during attraction entirely. The repulsion-shape comparison in Fig. 1(g) is equally clear-cut: UMAP is the sole method with large values at small distances (the shape is unbounded, though values are typically clipped during optimization). For additional details of these and related methods, see Appendix G.

To make these attraction/repulsion choices measurable, we embed MNIST while applying separate constant learning rates for attraction (λ_a) and repulsion (λ_r) and sweeping their values, scoring each embedding on two complementary metrics (Figs. 1(h,i)): trustworthiness (T) (Venna and Kaski 2001), a measure of structure preservation, and silhouette score (SIL) (Rousseeuw 1987), a measure of cluster separation. The sweep reveals a clean, asymmetric pattern for UMAP: both metrics peak only when $\lambda_a < 0.5$ (and consequently $\zeta_{-1} \simeq 0$), yet they are essentially insensitive to λ_r across the entire range $[0, 1]$. In other words, UMAP is fragile along the attraction axis but robust along the repulsion axis. NEG- t -SNE and PaCMAP, for which $\zeta_{-1} = 0$ by construction, remain effective across the full parameter range on both metrics. The comparison thus confirms directly that UMAP’s good performance hinges on driving ζ_{-1} close to 0—which it must accomplish through learning-rate annealing—whereas the newer algorithms obtain $\zeta_{-1} = 0$ for free. For additional discussions, results, and illustrations of these methods on different datasets, see Section 6 and Appendices B, G, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings depends heavily on proper initialization (Kobak and Linderman 2021; Wang et al. 2021). Typically, principal component analysis (PCA) of the data or spectral decomposition of the high-dimensional graph initializes the embedding, producing consistent mappings despite various sources of stochasticity.

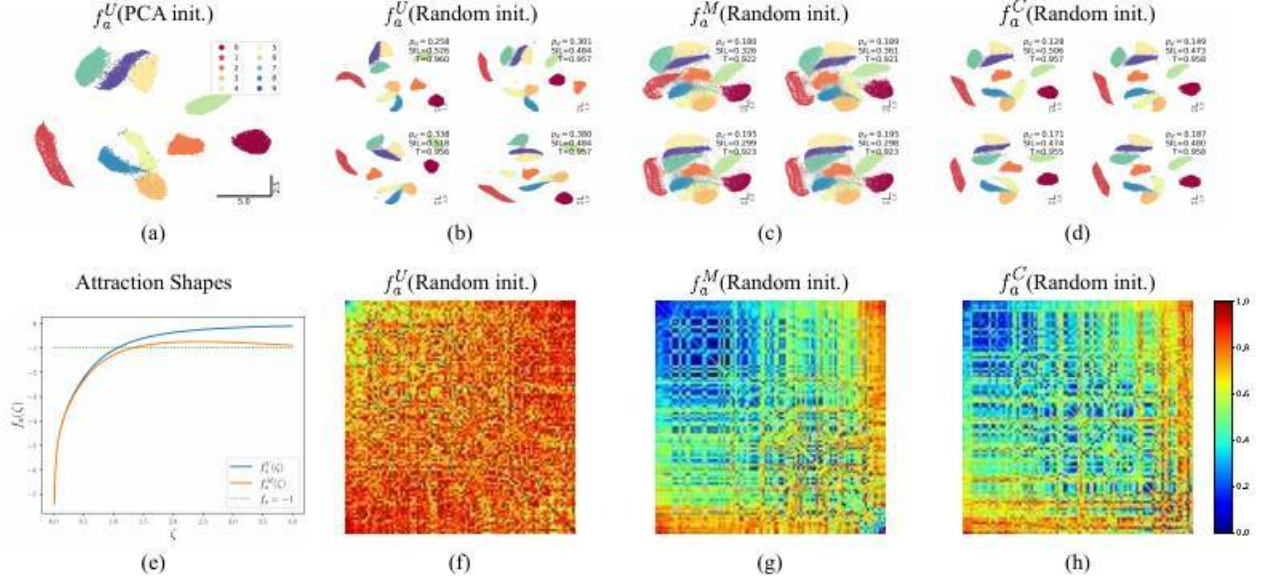


Figure 2: Random initialization on the MNIST dataset: modified attraction shapes cut the run-to-run Procrustes distance by roughly a third. (a) Mapping using PCA. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (mean $p_d = 0.78 \pm 0.13$), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes—a drop of 0.29 (37%) and 0.28 (36%), respectively, relative to UMAP. The diagonal (i, i) entries of the Procrustes matrix are sorted by p_d from (a), and the off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mapping. Read together, the lower mean p_d values and the larger similar (dark) regions in the matrices show that UMAP’s embeddings are not self-similar across runs, whereas both the modified and composite attraction shapes recover a consistent, scale-invariant structure.

Under random initialization, by contrast, clusters often fail to form or form in a different orientation on each run—a variability we quantify below with Procrustes distance. The mechanism is direct: if the initial distance between two nearest-neighbor points is large, the attractive force is too weak to draw them together. Known as near-sightedness (Wang et al., 2021), this failure is visible in the attraction shape itself, where $|f_a^U|$ decays to zero as distance grows (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

This diagnosis suggests a direct remedy: induce “far-sightedness” by increasing attraction at large distances so that faraway neighbors are still drawn together. To test this hypothesis quantitatively, we modify the attraction shape of UMAP to increase the attractive force:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β is a parameter that regulates the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). This addition in the attraction shape translates to adding a regularizer in the attractive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). In addition to attracting pairs at faraway distances, this technique enables intermixing of points that help convergence under random initialization (akin to early exaggeration in t -SNE). In (13), we chose the simplest linear correction; other functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), may also be suitable.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape attempts to remove any distortions introduced by f_a^M by reverting to the original UMAP. Below, we discuss the effects these modified and composite attraction shapes have on DR from random initialization.

We first created a PCA-initialized reference embedding of the MNIST dataset (Fig. 2(a)). We then produced randomly initialized (Gaussian) embeddings for each attraction shape and repeated the experiment 100 times per shape. To measure consistency, we use Procrustes analysis (Gower, 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1), aligning each randomly initialized embedding to the PCA-initialized reference and reporting their separation as the Procrustes distance (p_d)—lower is more consistent. Figure 2(b)

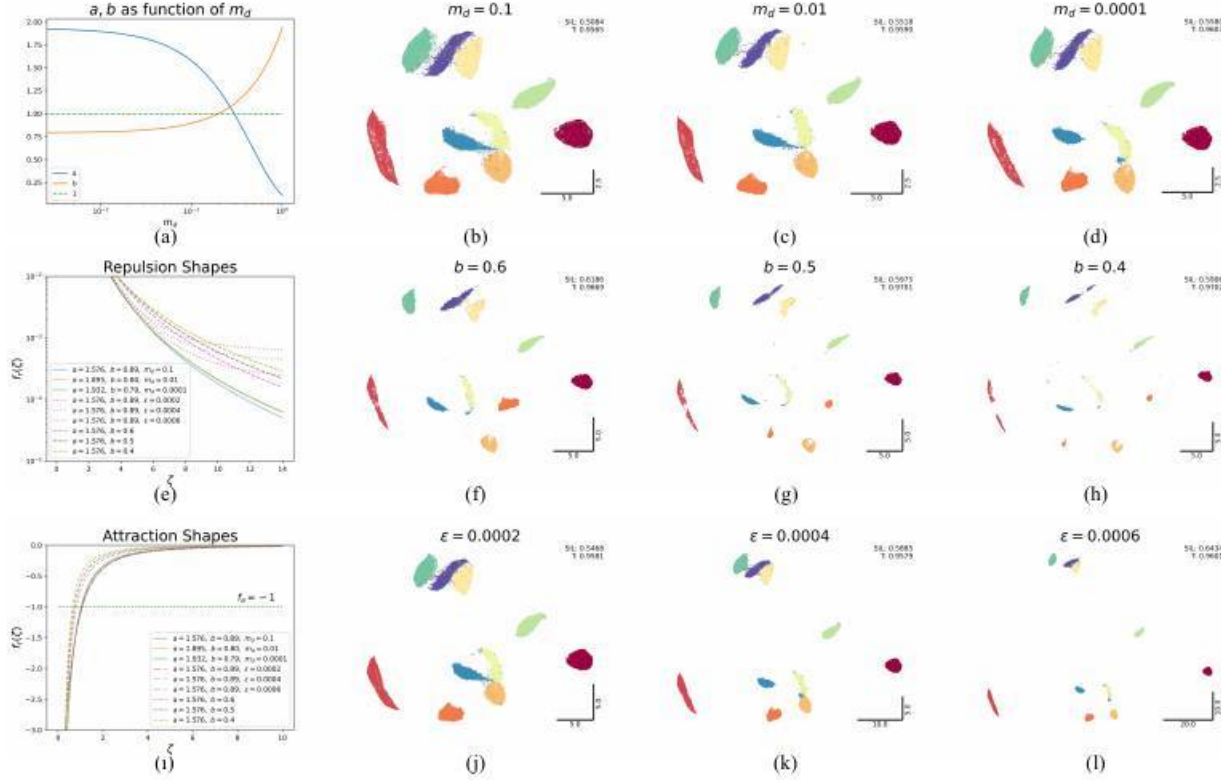


Figure 3: Control of inter-cluster distances on the MNIST dataset: varying m_d barely moves the repulsion shape, whereas varying b or adding ϵ visibly increases cluster separation. (a) Computing a, b by varying the low-dimensional distance m_d restricts the accessible parameter range. (b-d) UMAP output for $m_d = 0.1, 0.01$, and 0.0001 shows essentially no gain in cluster compactness across three orders of magnitude. (e) Repulsion shapes for different parameters. (f-h) Explicitly decreasing b instead yields progressively more compact clusters and splits off new ones that were previously absent. (i) Attraction shapes for the same parameter sweep. (j-l) Adding a small positive value (ϵ) to the repulsion shape steadily widens inter-cluster distance while leaving attraction untouched.

shows the four UMAP embeddings with the lowest p_d : even the best-matching runs preserve cluster shapes but disagree on placement. The modified and composite attraction shapes (Figs. 2(c) and (d)) visibly stabilize cluster placement, consistent with the one-third reduction in mean p_d reported in the caption.

To resolve the placements further, we consider the Procrustes matrix: its diagonal is sorted by p_d from the PCA-initialized mapping, and its off-diagonal values are p_d between two randomly initialized mappings (for details, see Appendix D.1); the construction is analogous to a similarity matrix (Foote 1999). The contrast is stark: default UMAP embeddings are mutually dissimilar (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes produce embeddings that are strongly similar to one another. Together with the lower mean p_d , this shows that UMAP encodes a unique structure independent of initialization, and that the default attraction shape simply falls short of reaching it because attraction is too weak at longer distances.

5.2 Cluster Formation and Compactness

The nominal control knob for cluster formation in UMAP is the minimum distance parameter m_d (through Eq. 2). We find, however, that it is a weak lever: varying m_d restricts the accessible values of a and b (Fig. 3(a)), so sweeping m_d across three orders of magnitude ($0.1, 0.01, 0.0001$) yields visually indistinguishable, non-improving embeddings (Figs. 3(b-d)). The reason is quantitatively clear: m_d barely perturbs the repulsion shape (Fig. 3(e)). Directly varying a and b is far more effective. Decreasing a does raise repulsion, but it suppresses attraction even faster (worsening near-sightedness), whereas decreasing b gives cleaner control (Figs. 3(e,i)). Stepping b down through $0.6, 0.5$, and 0.4 (Figs. 3(f-h)) monotonically increases inter-cluster distance and progressively breaks apart previously merged clusters. This break-up follows from the increasingly heavy-tailed kernel as b decreases (heavy-tailed kernels yield smaller, more distinct clusters [Van der Maaten and Hinton 2008, Kobak et al. 2019]; see Kobak et al. (2019) for how b modulates

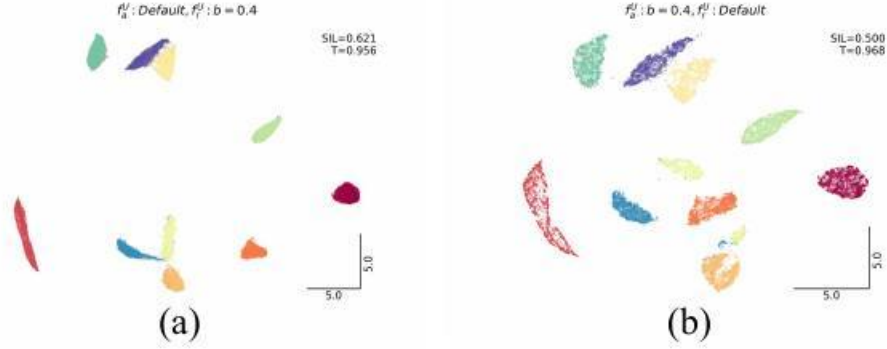


Figure 4: Isolating each force on the MNIST dataset by setting $b = 0.4$ in only one shape at a time. (a) Default attraction with $b = 0.4$ repulsion. (b) Default repulsion with $b = 0.4$ attraction. The two outcomes differ sharply: (a) reproduces exactly the same clusters as default UMAP, only more compact, whereas (b) develops new within-cluster structure and forms entirely new clusters. This one-force-at-a-time contrast pins cluster *creation* on attraction and cluster *compactness* on repulsion.

the tail in UMAP). This approach cleanly separates all ten MNIST labels, but it does not isolate the two forces: lowering b amplifies repulsion more strongly than m_d does, yet simultaneously reshapes the attraction profile—motivating the controlled, one-force-at-a-time experiments below.

To quantify the effect of repulsion independently, we can keep attraction fixed and vary the other. To achieve this, we modify the repulsion shape by adding a small positive value (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

while keeping the values of a and b constant, which effectively adds a regularizer in the repulsive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$). Using this, we can obtain stronger repulsion than previously (Fig. 3(e)). Figs. 3(j-l) show that inter-cluster distance grows monotonically with ε . Yet the resulting clustering closely mirrors that obtained by varying m_d —a loose separation of the labels—rather than the tighter, richer structure obtained by varying b . This is the telling comparison: because ε leaves the attraction shape untouched and varying m_d traces nearly the same attraction shapes (Fig. 3(i)), the two repulsion-only manipulations behave alike, indicating that cluster formation is governed predominantly by attraction.

To settle which force does what, we run the decisive controlled experiment: set $b = 0.4$ in exactly one shape while holding the other at its default (Fig. 4). With increased repulsion alone ($b = 0.4$ repulsion, default attraction), the embedding keeps the very same clusters as original UMAP but makes them smaller (Fig. 4(a))—no new structure appears. With increased attraction alone ($b = 0.4$ attraction, default repulsion), the embedding instead reveals additional within-cluster structure and even splits some existing clusters into smaller ones (Fig. 4(b)). The two conditions differ only in which force was strengthened, so the contrast is conclusive: attraction creates clusters, while repulsion compacts them (either by shrinking clusters or widening inter-cluster gaps, depending on the repulsion shape). Appendix G.3 shows that LocalMAP embeddings exploit this same attraction-repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Setting a and b to 1 is common across dimensionality reduction algorithms because it makes the gradients bounded and thereby eases optimization. Recently, Damrich et al. (2023) exploited this for UMAP and proposed NEG- t -SNE, which not only stabilizes the gradient but also yields more compact clustering even at a constant learning rate of 1—the regime in which UMAP itself degrades. In parallel, Parametric UMAP (Sainburg et al. 2021) began with the UMAP loss but later adopted a numerically stable modified cross-entropy loss (Shi et al. 2023) with a logsigmoid kernel. Comparing the two reveals that they reach the same formulation from different starting points, which we state precisely as:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. Even with $a = b = 1$, the attraction shape is bounded within $[-2, 0]$ with $f_a^U(0) = -2$

github.com, merged on April 26, 2022

(Fig. 1(f)), while the repulsion shape is essentially unchanged (still unbounded as $\zeta \rightarrow 0$, Fig. 1(g)). Because f_a^U still drops below -1 as $\zeta \rightarrow 0$, Proposition 4.1 and the discussion in Section 5 imply that this unity case continues to require learning-rate annealing—the same failure mode as default UMAP, and precisely the one NEG- t -SNE avoids.

Damrich et al. (2023) compared UMAP’s negative sampling loss function from the perspective of contrastive embedding (CE) and concluded that the effective kernel of UMAP is $1/\zeta^2$. Under the CE framework, the authors introduced NEG- t -SNE by changing the kernel to $1/(1 + \zeta^2)$. Reverting to UMAP formalism, this results in the low-dimensional affinity function $q_{ij}^N = 1/(2 + \zeta^2)$. Consequently, the attraction and repulsion shapes are $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). Here the comparison turns decisively in NEG- t -SNE’s favor: its attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1 so any $\lambda \in [0, 1]$ contracts pairs without the expansion-contraction oscillation that afflicts UMAP. Consequently NEG- t -SNE is far less sensitive to learning-rate annealing, and its clusters stay sharp even at a constant $\lambda = 1.0$. Its repulsion shape is likewise bounded within $[0, 1]$ and, unlike UMAP’s, does not blow up as $\zeta \rightarrow 0$, so fewer points are ejected from clusters when sampled. Whereas Damrich et al. (2023) attributed this advantage solely to the bounded repulsive forces of NEG- t -SNE versus the unbounded ones of UMAP, our side-by-side shape analysis (Fig. 1) shows the attraction shape is equally responsible for the observed stability and compactness. For additional discussion on NEG- t -SNE with illustration, see Appendix F. Together with the discussion in Fig. 4 this suggests that attraction and repulsion shapes from different methods can be mixed, for instance, using the attraction shape of UMAP with the repulsion shape of NEG- t -SNE. We show a few examples by experimenting on UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work, we quantified how attractive and repulsive forces jointly drive cluster formation. The coarse picture—attraction pulls similar points together, repulsion pushes dissimilar points apart—was already known, but the exact mechanisms were not, and could not be attributed to a specific force under controlled comparison. Through one-force-at-a-time experiments and shape-level comparisons across UMAP, NEG- t -SNE, and PaCMAP, we made those attributions concrete and measurable. While the main text centers on UMAP and NEG- t -SNE, the same shape framework and comparisons extend to other dimensionality reduction methods (Appendix G).

We treated the attraction and repulsion coefficients as functional mappings and analyzed their shapes. Characterizing these shapes (Propositions 4.1 and 4.2) exposed a counterintuitive, quantitatively precise result: below the threshold distance ζ_{-1} , UMAP’s attractive force actually expands distance rather than contracting it (Fig. 1(b)), pushing near neighbors apart instead of pulling them together. We show that UMAP’s learning-rate schedule works precisely by driving this threshold ζ_{-1} toward zero.

We also studied initialization of the low-dimensional embedding. Similar points that start far apart feel too little attraction and never contract adequately. Our formalism offers a direct fix—adding a term to the attraction shape to strengthen attraction at large distances—and the effect is measurable: across MNIST, FMNIST, and single-cell transcriptomes, the modified and composite shapes cut the mean Procrustes distance by roughly one-third or more (e.g., $0.78 \rightarrow 0.49$ on MNIST, $0.72 \rightarrow 0.38$ on FMNIST, $0.91 \rightarrow 0.61$ on single-cell data), sharply improving consistency under random initialization. This gives us confidence that UMAP encodes a unique structure, with numerical tricks merely helping reach it faster.

Analyzing the repulsion shape showed quantitatively that higher repulsion produces larger inter-cluster distance. Adding a small positive constant to the shape delivers a level of repulsion that varying the minimum-distance parameter m_d cannot reach, even across three orders of magnitude of m_d .

Finally, we compared NEG- t -SNE and UMAP directly. For $a = 1$ and $b = 1$, the decisive difference is that NEG- t -SNE keeps both attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) bounded, so both propositions hold for every learning rate in $[0, 1]$; NEG- t -SNE is therefore markedly less sensitive to a fixed learning rate, whereas UMAP is not. Varying a and b in NEG- t -SNE reintroduces the same numerical challenges as UMAP. We also proved that NEG- t -SNE and Parametric UMAP are equivalent.

These quantitatively grounded insights into attraction–repulsion dynamics offer concrete tools for optimizing dimensionality reduction algorithms. Beyond this, the close connection between dimensionality reduction and contrastive learning (Damrich and Hamprecht 2021, Hu et al. 2023) suggests our approach can also enhance representation learning. Taken together, and supported by consistent gains measured across three datasets and multiple metrics, our work makes embeddings and their interpretations more principled, more consistent, and more reliable, and charts clear directions for future research.

Software and Data

All the data used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com
 Fashion-MNIST is available at github.com
 Single-cell transcriptomes data is available at github.com

References

- Akshay Agrawal, Alnur Ali, Stephen Boyd, et al. Minimum-distortion embedding. *Foundations and Trends® in Machine Learning*, 14(3):211–378, 2021.
- Mahsun Altin and Altan Cakir. Exploring the influence of dimensionality reduction on anomaly detection performance in multivariate time series. *IEEE Access*, 2024.
- Ehsan Amid and Manfred K Warmuth. TriMap: Large-scale dimensionality reduction using triplets. *arXiv preprint arXiv:1910.00204*, 2019.
- Sanjeev Arora, Wei Hu, and Pravesh K Kothari. An analysis of the t-sne algorithm for data visualization. In *Conference On Learning Theory*, pages 1455–1462. PMLR, 2018.
- Etienne Becht, Leland McInnes, John Healy, Charles-Antoine Dutertre, Immanuel WH Kwok, Lai Guan Ng, Florent Ginhoux, and Evan W Newell. Dimensionality reduction for visualizing single-cell data using umap. *Nature biotechnology*, 37(1):38–44, 2019.
- Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps and spectral techniques for embedding and clustering. In *Advances in neural information processing systems*, pages 585–591, 2002.
- Jan Niklas Böhm, Philipp Berens, and Dmitry Kobak. Attraction-repulsion spectrum in neighbor embeddings. *The Journal of Machine Learning Research*, 23(1):4118–4149, 2022.
- Ingwer Borg and Patrick JF Groenen. *Modern multidimensional scaling: Theory and applications*. Springer Science & Business Media, 2007.
- T Tony Cai and Rong Ma. Theoretical foundations of t-sne for visualizing high-dimensional clustered data. *Journal of Machine Learning Research*, 23(301):1–54, 2022.
- Tara Chari and Lior Pachter. The specious art of single-cell genomics. *PLOS Computational Biology*, 19(8):e1011288, 2023.
- Sebastian Damrich and Fred A Hamprecht. On UMAP’s true loss function. *Advances in Neural Information Processing Systems*, 34:5798–5809, 2021.
- Sebastian Damrich, Niklas Böhm, Fred A Hamprecht, and Dmitry Kobak. From t-SNE to UMAP with contrastive learning. In *The Eleventh International Conference on Learning Representations*, 2023.
- Cyril de Bodt, Alex Diaz-Papkovich, Michael Bleher, Kerstin Bunte, Corinna Coupette, Sebastian Damrich, Enrique Fita Sanmartin, Fred A Hamprecht, Emőke-Ágnes Horvát, Dhruv Kohli, et al. Low-dimensional embeddings of high-dimensional data. *arXiv preprint arXiv:2508.15929*, 2025.
- Andrew Draganov and Simon Dohn. Relating tsne and umap to classical dimensionality reduction. *arXiv e-prints*, pages arXiv–2306, 2023.
- Andrew Draganov, Jakob Jørgensen, Katrine Scheel, Davide Mottin, Ira Assent, Tyrus Berry, and Cigdem Aslay. ActUp: Analyzing and consolidating tSNE and UMAP. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pages 3651–3658, 8 2023.
- Shu-Kai S Fan, Du-Ming Tsai, Chih-Hung Jen, Chia-Yu Hsu, Fei He, and Li-Ting Juan. Data visualization of anomaly detection in semiconductor processing tools. *IEEE Transactions on Semiconductor Manufacturing*, 35(2):186–197, 2021.
- Jason Fleischer and Mohammad Tariqul Islam. Late breaking abstract-identifying and phenotyping COVID-19 patients using machine learning on chest x-rays. *European Respiratory Journal*, 2020.
- Jonathan Foote. Visualizing music and audio using self-similarity. In *Proceedings of the seventh ACM international conference on Multimedia (Part 1)*, pages 77–80, 1999.
- Rita González-Márquez, Luca Schmidt, Benjamin M Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 2024.
- John C Gower. Generalized procrustes analysis. *Psychometrika*, 40:33–51, 1975.

- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(1):1–23, 2021.
- Geoffrey Hinton and Sam T Roweis. Stochastic neighbor embedding. In *Advances in Neural Information Processing Systems*, volume 15, pages 833–840, 2002.
- Tianyang Hu, Zhili Liu, Fengwei Zhou, Wenjia Wang, and Weiran Huang. Your contrastive learning is secretly doing stochastic neighbor embedding. In *The Eleventh International Conference on Learning Representations*, 2023.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, pages 20617–20642, 2024.
- Mohammad Tariqul Islam and Jason W Fleischer. Manifold-aligned neighbor embedding. In *ICLR 2022 Workshop on Geometrical and Topological Representation Learning*, 2022.
- Mohammad Tariqul Islam and Jason W Fleischer. Outlier detection in large radiological datasets using umap. In *International Workshop on Topology-and Graph-Informed Imaging Informatics*, pages 111–121. Springer, 2024.
- Robert A Jacobs. Increased rates of convergence through learning rate adaptation. *Neural networks*, 1(4):295–307, 1988.
- Mathieu Jacomy, Tommaso Venturini, Sebastien Heymann, and Mathieu Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the gephi software. *PloS one*, 9(6):e98679, 2014.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- Dmitry Kobak and George C Linderman. Initialization is critical for preserving global data structure in both t-sne and umap. *Nature biotechnology*, 39(2):156–157, 2021.
- Dmitry Kobak, George Linderman, Stefan Steinerberger, Yuval Kluger, and Philipp Berens. Heavy-tailed kernels reveal a finer cluster structure in t-sne visualisations. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 124–139. Springer, 2019.
- Alexander Kolpakov and Aidan Rocke. The information geometry of umap. *Le Matematiche*, 79(1):151–164, 2024.
- Nikita Kotlov, Kirill Shaposhnikov, Cagdas Tazearslan, Madison Chasse, Artur Baisangurov, Svetlana Podsvirova, Dawn Fernandez, Mary Abdou, Leznath Kaneunyenye, Kelley Morgan, et al. Procrustes is a machine-learning approach that removes cross-platform batch effects from clinical rna sequencing data. *Communications Biology*, 7(1):392, 2024.
- Joseph B Kruskal. Nonmetric multidimensional scaling: a numerical method. *Psychometrika*, 29(2):115–129, 1964.
- Noël Kury, Dmitry Kobak, and Sebastian Damrich. DREAMS: Preserving both local and global structure in dimensionality reduction. *arXiv preprint arXiv:2508.13747*, 2025.
- Siu Kwan Lam, Antoine Pitrou, and Stanley Seibert. Numba: A llvm-based python jit compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015.
- Pierre Lambert, Cyril De Bodt, Michel Verleysen, and John A Lee. Squadmds: a lean stochastic quartet mds improving global structure preservation in neighbor embedding like t-sne and umap. *Neurocomputing*, 503:17–27, 2022.
- George C Linderman and Stefan Steinerberger. Clustering with t-SNE, provably. *SIAM Journal on Mathematics of Data Science*, 1(2):313–332, 2019.
- George C Linderman and Stefan Steinerberger. Dimensionality reduction via dynamical systems: The case of t-sne. *SIAM Review*, 64(1):153–178, 2022.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9(Nov):2579–2605, 2008.
- Evan Z Macosko, Anindita Basu, Rahul Satija, James Nemesh, Karthik Shekhar, Melissa Goldman, Itay Tirosh, Allison R Bialas, Nolan Kamitaki, Emily M Martersteck, et al. Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell*, 161(5):1202–1214, 2015.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, pages 3111–3119, 2013.
- Ashwin Narayan, Bonnie Berger, and Hyunghoon Cho. Assessing single-cell transcriptomic variability through density-preserving data visualization. *Nature Biotechnology*, 39(6):765–774, 2021.

- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Pavlin G. Poličar, Martin Stražar, and Blaž Zupan. opentsne: A modular python library for t-sne dimensionality reduction and embedding. *Journal of Statistical Software*, 109(3):1–30, 2024.
- Aditya Ravuri and Neil D Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne. *arXiv preprint arXiv:2405.17412*, 2024.
- Aditya Ravuri, Francisco Vargas, Vidhi Lalchand, and Neil D Lawrence. Dimensionality reduction as probabilistic inference. In *ICML 2023 Workshop on Structured Probabilistic Inference* & *Generative Modeling*, 2023.
- Peter J Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20:53–65, 1987.
- Sam T Roweis and Lawrence K Saul. Nonlinear dimensionality reduction by locally linear embedding. *science*, 290(5500):2323–2326, 2000.
- Tim Sainburg, Leland McInnes, and Timothy Q Gentner. Parametric umap embeddings for representation and semisupervised learning. *Neural Computation*, 33(11):2881–2907, 2021.
- Uri Shaham and Stefan Steinerberger. Stochastic neighbor embedding separates well-separated clusters. *arXiv preprint arXiv:1702.02670*, 2017.
- Xintong Shi, Wenzhi Cao, and Sebastian Raschka. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26(3):941–955, 2023.
- Jian Tang, Jingzhou Liu, Ming Zhang, and Qiaozhu Mei. Visualizing large-scale and high-dimensional data. In *Proceedings of the 25th international conference on world wide web*, pages 287–297, 2016.
- Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9(11), 2008.
- Jarkko Venna and Samuel Kaski. Neighborhood preservation in nonlinear projection methods: An experimental study. In *International Conference on Artificial Neural Networks*, pages 485–491. Springer, 2001.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Yingfan Wang, Haiyang Huang, Cynthia Rudin, and Yaron Shaposhnik. Understanding how dimension reduction tools work: an empirical approach to deciphering t-SNE, UMAP, TriMAP, and PaCMAP for data visualization. *The Journal of Machine Learning Research*, 22(1):9129–9201, 2021.
- Yingfan Wang, Yiyang Sun, Haiyang Huang, and Cynthia Rudin. Dimension reduction with locally adjusted graphs. In *Thirty-Ninth AAAI Conference on Artificial Intelligence*, 2025.
- Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. 2017.
- Yang Yang, Hongjian Sun, Jialei Gong, Yali Du, and Di Yu. Interpretable dimensionality reduction by feature preserving manifold approximation and projection. *arXiv preprint arXiv:2211.09321*, 2022.
- Lihi Zelnik-Manor and Pietro Perona. Self-tuning spectral clustering. *Advances in neural information processing systems*, 17, 2004.
- Jonathan X Zheng, Samraat Pawar, and Dan FM Goodman. Graph drawing by stochastic gradient descent. *IEEE transactions on visualization and computer graphics*, 25(9):2738–2748, 2018.

Appendix

In Appendix [A](#) we provide necessary derivations regarding attraction and repulsion shapes of UMAP. Following our discussion in the main text, we explore usage of a constant learning rate again in Appendix [B](#). Then, in Appendix [C](#) we discuss the implications of improved attraction and repulsion on large language model embeddings. We formally define Procrustes distance and matrix in Appendix [D.1](#). We explore additional datasets for the random initialization experiment in Appendix [E](#). We extend the discussion of NEG- t -SNE and UMAP in Appendix [F](#). After that, in Appendix [G](#) we explore alternate dimensionality reduction methods. Then, for completion, we discuss the construction of the high-dimensional graph in these methods in Appendix [H](#). In Appendix [I](#) we show the detailed result of varying λ_a and λ_r (from Fig. [1](#)) for UMAP, NEG- t -SNE, and PaCMAP on different datasets. We show the effect of mixing and matching the shapes from different algorithms in Appendix [J](#). Finally, we provide implementation details in Appendix [K](#).

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

We use a general form of the low-dimensional affinity function, i.e., $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, to derive the attraction shape. It reduces to UMAP for $\gamma = 1$ and to NEG- t -SNE for $\gamma = 2$. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

Its behavior with distance can be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

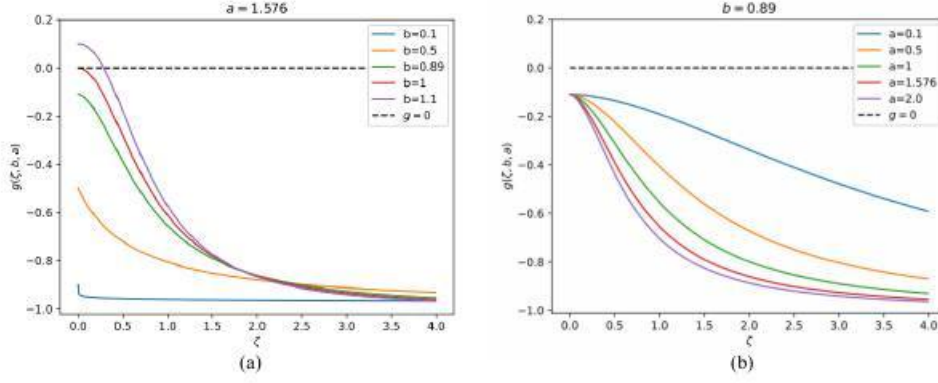


Figure 5: Values of $g(\zeta, b, a)$ for (a) a fixed at 1.576 and (b) b fixed at 0.89.

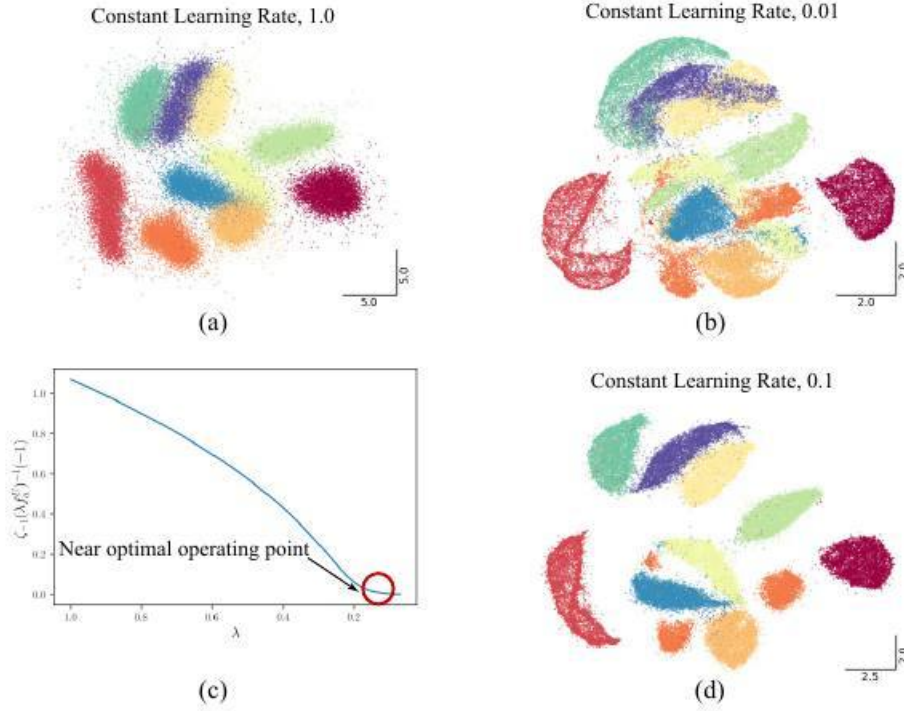


Figure 6: Constant learning rate traces a clear trade-off in the embeddings. (a) Too high ($\lambda = 1.0$): embeddings are diffuse, because ζ_{-1} is large. (b) Too low ($\lambda = 0.01$): no clusters form, because attraction and repulsion are both too weak. (c) ζ_{-1} falls nonlinearly as λ decreases, so the objective is to shrink ζ_{-1} while retaining effective force strength. (d) At the near-optimal constant $\lambda = 0.1$, this balance is struck and distinct, compact clusters with clear boundaries emerge.

This leads to a strictly increasing condition ($\frac{df^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality is valid as long as $0 < b \leq 1$ (using the derivative and asymptotes of g). Figure 5 shows values of g for different b and a . As b increases above 1, the inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The first term gives the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsive term due to Eq. (15) is given by

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the additional term acts as a regularizer, simply using norms. However, when we directly add this term to attraction and repulsion shapes, we can easily explain what each term is doing. For the attractive term, it is increasing far-sightedness, whereas for the repulsive term, it adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

As discussed in Sections 5 and 6 UMAP is believed to require learning-rate annealing (Fig 6). Section 5 explained this via the minimum distance for contraction (ζ_{-1}): driving ζ_{-1} down through annealing is what produces compact clusters. Section 6 then compared the attraction shapes of UMAP ($a = 1.0$, $b = 1.0$) and NEG- t -SNE, and the comparison is quantitatively clean: NEG- t -SNE tolerates a constant $\lambda = 1.0$ far better than UMAP because its attraction shape lies in $[-1, 0]$ whereas UMAP’s spans $[-2, 0]$. The same logic predicts a fix for UMAP: constraining its attraction shape to $[-1, 0]$ (via $\lambda = 0.5$) lets it withstand a constant learning rate, which the embeddings in Figs. 11 f,h) confirm.

Even so, annealing still yields the best embeddings across a wide range of a and b . The reason is quantitative: the objective is ultimately to drive ζ_{-1} to zero, which happens only as λ approaches zero. Fixing λ too high leaves ζ_{-1} large and the embedding diffuse (Fig 6 a)), whereas fixing it too low starves both forces so that no clear clusters form (Fig 6 b)). The ζ_{-1} -versus- λ curve (Fig 6 c)) identifies $\lambda = 0.1$ as a sweet spot—low ζ_{-1} with still-substantial attraction and repulsion—and at this constant $\lambda = 0.1$ we indeed recover compact clusters with clear boundaries (Fig 6 d)).

C Large Language Model Embedding

Increased attraction and repulsion produce measurably better embeddings, and the effect scales to large language model features. To show this, we drew 300,000 PubMedBERT (Gu et al. 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP yields a crowded structure in which the label colors overlap (Fig. 7 a)); absent the labels, the embedding would look structureless, as no explicit clusters are visible. Reducing b to 0.6 (Section 5.2) raises repulsion while simultaneously strengthening attraction, and the mapping becomes explorable even without labels (Fig. 7 b)). The gain is quantitative as well as visual: on a 10,000-point hold-out set, 10-NN accuracy rises from 49.2% to 55.48%—an absolute improvement of 6.28 points—simply by lowering b to 0.6.

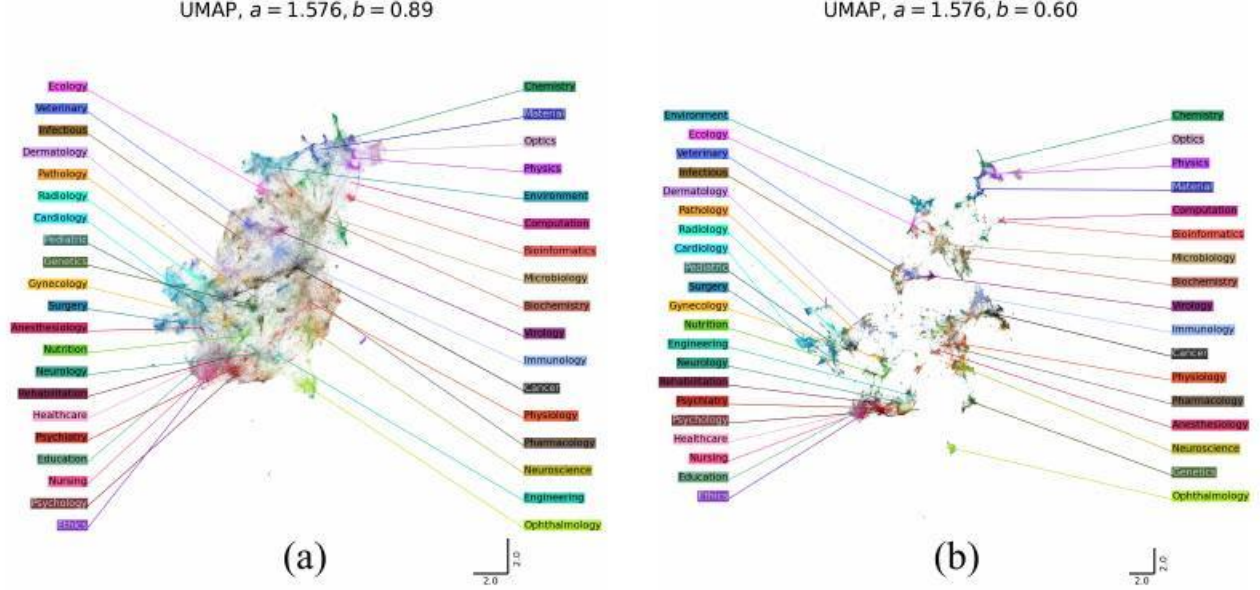


Figure 7: Visualization of 300K abstracts from the PubMed dataset: reducing b from 0.89 to 0.60 turns overlapping labels into well-separated clusters. (a) Default UMAP ($a = 1.576, b = 0.89$) crowds the labels (colors) on top of one another. (b) UMAP with improved repulsion and attraction ($a = 1.576, b = 0.60$) forms cleaner clusters with non-overlapping labels, and lifts hold-out 10-NN accuracy from 49.2% to 55.48% (a 6.28-point gain; see text). Both mappings are labeled for easier reading.

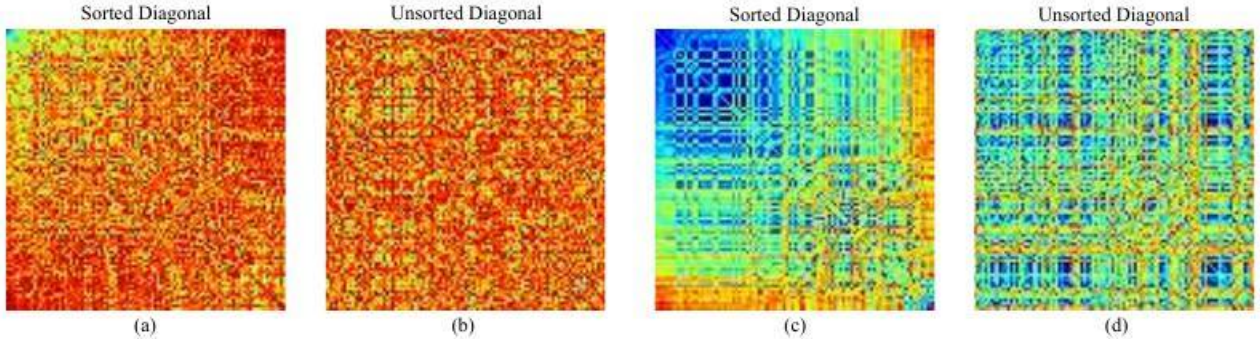


Figure 8: Sorting the Procrustes-matrix diagonal preserves the numbers but reveals the self-similarity visually. (a) Matrix from [2]f), default UMAP attraction shape, with the diagonal sorted by p_d against a PCA-initialized sample. (b) Identical data to (a) but with the diagonal unsorted. (c) Matrix from [2]g), modified attraction shape, diagonal sorted the same way—note the visibly larger block of mutually similar embeddings. (d) Identical data to (c) but unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower 1975) measures similarity between two point clouds $\{x\}$ and $\{y\}$ under linear transformations, viz. translation, scaling, and rotation. Operationally, we hold the former fixed and vary the latter until the two sets are in maximum alignment. Let $\{y'\}$ be the transformation of $\{y\}$ that achieves this objective. Then the Procrustes distance is given by

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proven useful in a variety of settings (McInnes et al. 2018; Islam and Fleischer 2022; Kotlov et al. 2024).

Here, we use the Procrustes distance to measure the consistency of embedding under random initialization. Let $\{x\}_p$ be a reference embedding (using PCA initialization in our experiments), and $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be a set of N embeddings obtained from random initialization. The similarity of the embeddings can be quantified by taking a mean and standard deviation of the strictly lower triangular values of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indexes of X_r can be sorted such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, so that the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal leaves the reported mean and standard deviation unchanged, but it makes the underlying self-similarity of the embeddings visually legible. In Fig. 8(a), the sorted diagonal makes similar embeddings clump in the upper-left region of the matrix. With the modified attraction shape, that region of mutually similar embeddings grows substantially (the larger blue block in Fig. 8(c)), signaling a metastable point that many random runs converge to—directly consistent with the lower mean p_d reported earlier. Without sorting, the block dissolves and this evidence of similarity is no longer apparent (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. Higher T (up to 1) means better preservation of local neighborhoods. We set $k = 5$, assessing preservation of each point’s five nearest neighbors. For efficiency we compute T on a random sample of 10,000 indices, and—critically for fair comparison—reuse the exact same indices across all embeddings being compared.

D.3 Silhouette Score

The silhouette score (Rousseeuw, 1987) is normally used to assess clustering algorithms; here we repurpose it as a direct measure of label separation—how well the embedding groups points sharing a ground-truth label, on the premise that a good embedding should separate the labels as much as possible. For a point y_i in a point cloud $\{y\}$, let a_i be the mean distance from y_i to other points of its own label and b_i the minimal mean distance from y_i to points of any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies within $[-1, 1]$, with higher being better: near 1 means y_i sits comfortably within its own label cluster, near 0 marks a boundary point, and near -1 signals failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

Unlike trustworthiness, we computed the silhouette score over the whole embedding (no random sampling) using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reported the random-initialization experiment on MNIST. We repeat it here on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Figs. 9 and 10). The conclusion holds and the margins are, if anything, larger: increasing attraction at large distances (modified and composite shapes) cuts the mean Procrustes distance from 0.72 to 0.38 on FMNIST (a 47% reduction) and from 0.91 to 0.61 on single-cell data (a 33% reduction), mirroring the MNIST result and confirming that the consistency gain transfers across datasets.

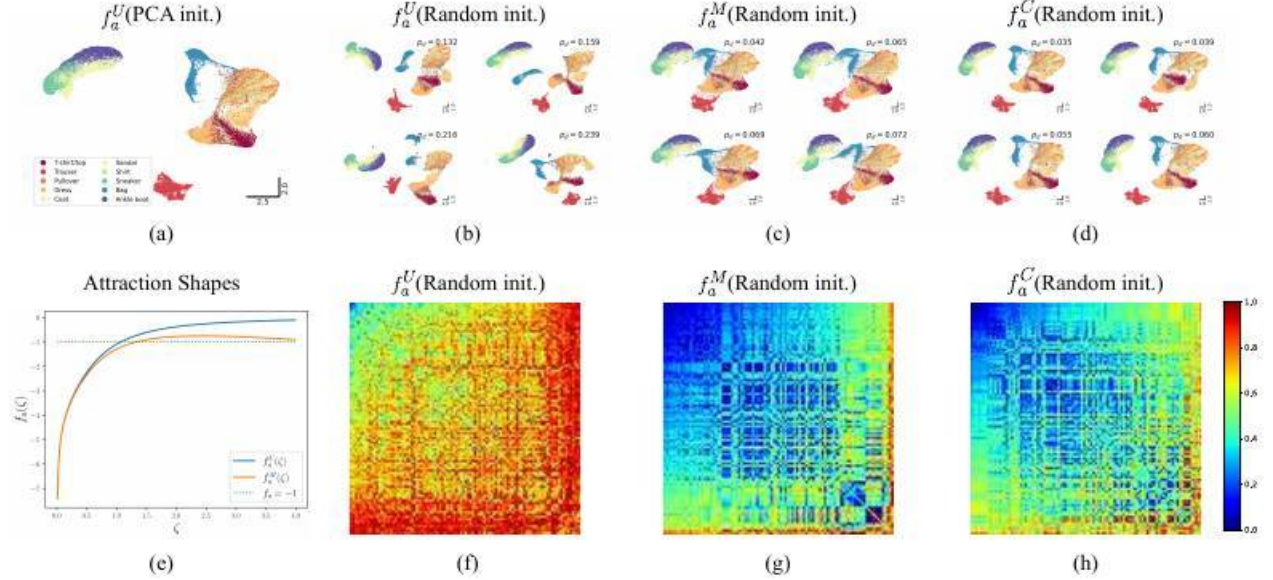


Figure 9: Random UMAP initialization on FMNIST: modified attraction nearly halves the run-to-run Procrustes distance. (a) PCA reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) default (mean $p_d = 0.72 \pm 0.15$), (g) modified (0.38 ± 0.19 , a 47% reduction), and (h) composite (0.43 ± 0.18 , a 40% reduction) attraction shapes.

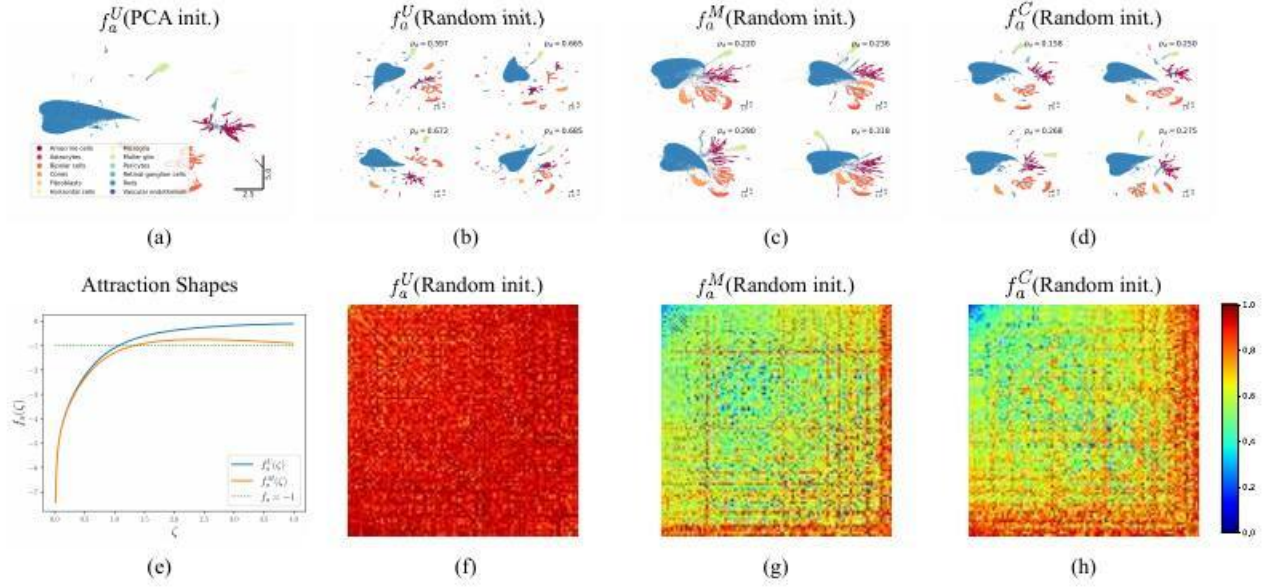


Figure 10: Random UMAP initialization on single-cell transcriptomes: the hardest case for default UMAP shows the clearest gain. (a) PCA reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) default (mean $p_d = 0.91 \pm 0.06$, the least consistent of all datasets), (g) modified (0.61 ± 0.13 , a 33% reduction), and (h) composite (0.64 ± 0.15 , a 30% reduction) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

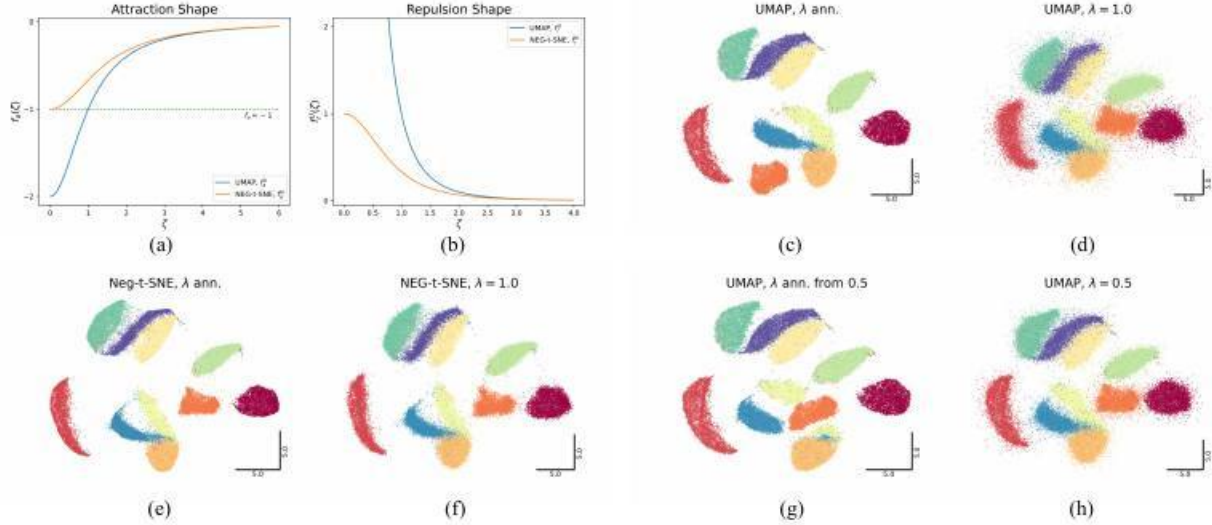


Figure 11: Learning-rate sensitivity of UMAP versus NEG- t -SNE on MNIST, side by side. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1, b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to λ because $f_a^U < -1$ as the separation ζ shrinks, so without annealing (d) the clusters turn fuzzy. (e,f) NEG- t -SNE is much less sensitive because $f_a^N \in [-1, 0]$ always, keeping clusters sharp even un-annealed. (g,h) Constraining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ recovers the same insensitivity for UMAP—directly demonstrating that the attraction range, not the method identity, drives the difference.

Figure 11 places the shapes of UMAP and NEG- t -SNE alongside their MNIST embeddings. Under annealing the two methods look alike (Figs. 11(c,e)), so the difference is invisible in that regime. The gap opens only at constant $\lambda = 1$: UMAP produces a fuzzy structure while NEG- t -SNE keeps markedly cleaner boundaries (Fig. 11(d,f)). Section 6 predicts that constraining f_a^U to $[-1, 0]$ should sharpen UMAP’s clusters at fixed λ —a prediction already foreshadowed in Fig. 1(h,i), where UMAP scored best on both metrics once $\lambda_a \lesssim 0.5$. Setting $\lambda = 0.5$ (which satisfies Proposition 4.1 for all ζ) tests this directly. The resulting embeddings (Figs. 11(g,h)) confirm the prediction quantitatively: UMAP’s clusters now resemble NEG- t -SNE’s and are visibly sharper than at $\lambda = 1$ (Fig. 11(d,h)). A few points still stray outside the clusters, a residual effect of UMAP’s unbounded repulsion shape that NEG- t -SNE resolves.

Next, we can introduce the parameters a and b into NEG- t -SNE (essentially the formulation of Parametric UMAP; see Proposition 6.1). The affinity function becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes again become unbounded as $\zeta \rightarrow 0$, so NEG- t -SNE inherits exactly the same numerical challenges as UMAP once a and b are allowed to vary—the stability advantage is specific to the $a = b = 1$ setting. One measurable distinction remains: relative to UMAP, this attraction shape reaches a lower minimum distance ζ_{-1} . This can aid cluster formation, but the shape also decays to zero faster (stronger near-sightedness at large distances), trading away some effectiveness of attraction over long ranges.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al., 2021), but later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). This modification makes Parametric UMAP and Neg- t -SNE (Damrich et al., 2023) equivalent (Proposition 6.1). We show the equivalence below.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

To make cross-method comparison exact, the alternative algorithms we consider all share UMAP's kernel function (with $a = 1$ and $b = 1$ in their low-dimensional weight):

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

In this section, we first discuss the TriMap (Amid and Warmuth 2019) algorithm. Even though, this algorithm relies on triplets (and not pairwise interactions), this works as a primer for analyzing attraction and repulsion that are a bit more involved than UMAP. This discussion is followed by Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. 2021) and its extension Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al. 2025) that modify TriMAP's loss function that works for pairwise interactions. We then tackle t -SNE (Van der Maaten and Hinton 2008). Then, we provide a short note on SNE (Hinton and Roweis 2002) that uses an alternate kernel function and finally end the section by briefly discussing multidimensional scaling (Borg and Groenen 2007).

G.1 TriMap

TriMap (Amid and Warmuth 2019) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j is in the k -nearest neighbor set of y_i in the high dimension, and y_k is a far-away point. When minimized, we expect that y_i and y_j attract each other, while y_i and y_k repel each other. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ is the distance from the faraway point, and f_a^T and f_r^T are attraction and repulsion shapes of TriMap, respectively. Unlike UMAP, the attractive and repulsive components are non-separable and the shapes depend on two distance measures (making them 2D). The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

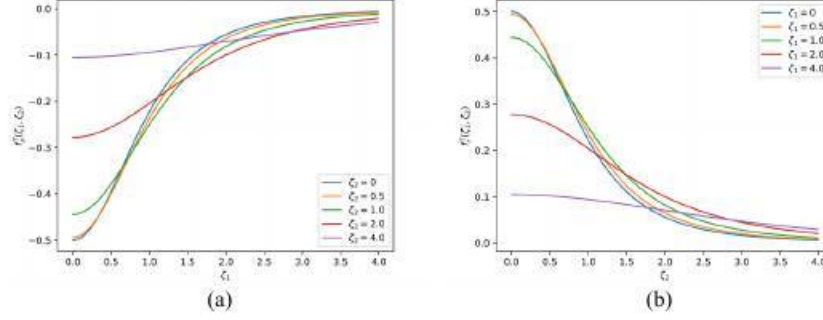


Figure 12: (a) Attraction shapes of TriMap for different ζ_2 and (b) repulsion shapes of TriMap for different ζ_1 .

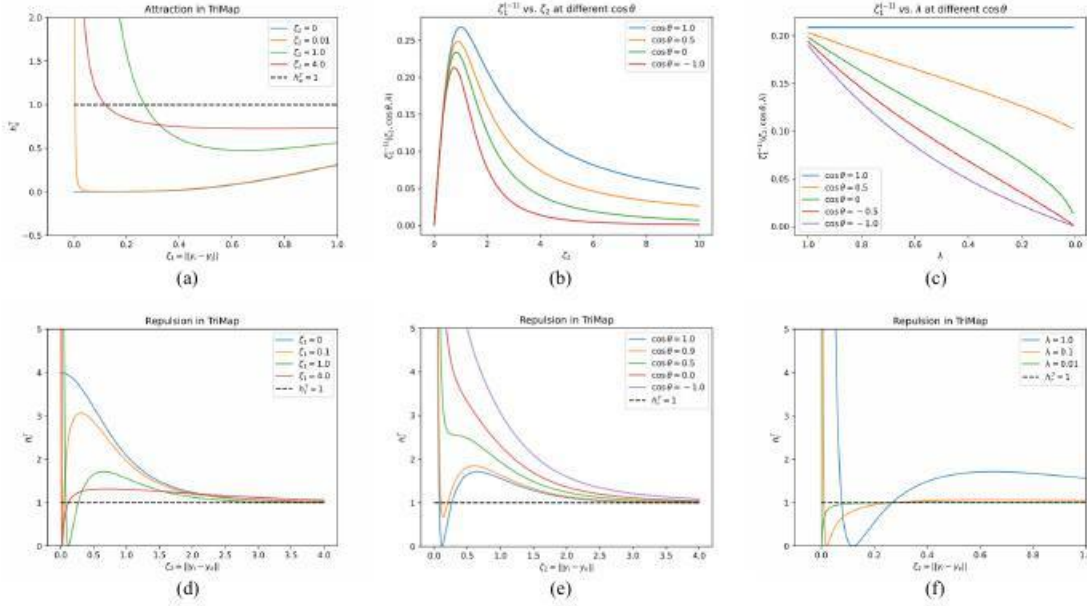


Figure 13: Attraction and repulsion behavior in TriMap, contrasted point-by-point with UMAP. (a) h_a^T vs ζ_1 for different ζ_2 ; values below the dotted line indicate attraction. As in UMAP, nearest neighbors (y_i, y_j) experience both attraction and repulsion. (b,c) The key difference from UMAP: the minimum distance for contraction ($\zeta_1^{(-1)}$) is no longer a single number but varies with (b) ζ_2 and (c) λ , with $\cos \theta$ setting its range. (d-f) Repulsion in TriMap while varying (d) ζ_1 , (e) $\cos \theta$, and (f) λ ; values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsion is purely repulsive, TriMap's can act in either direction. Unless labeled otherwise, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap’s attraction and repulsion shapes (Fig. 12) follow trends similar to UMAP’s, with one quantitative difference in TriMap’s favor: its repulsion shape bottoms out at -0.5 , so—unlike UMAP—attraction alone never flips a pair’s position. Because Eqs. (45)-(47) do not decouple the attractive and repulsive terms, Propositions 4.1 and 4.2 no longer apply directly. Treating attraction first, we show

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i - y_j)^T (y_i - y_k)}{\|y_i - y_j\|_2 \|y_i - y_k\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$ and λ . In particular, the value of $\zeta_1 = \|y_i - y_j\|_2$, where we want a contraction, is coupled with additional variables. Figure 13(a) shows the attraction behavior for various values of ζ_2 , while $\cos \theta = 1$ and $\lambda = 1$. The values below the dotted line indicate attraction (and thus contraction of distance ζ_1), whereas the values above indicate repulsion (and therefore expansion of distance ζ_1). The value where the dotted line and h_a^T meet gives the minimum distance for contraction ($\zeta_1^{(-1)}$) [analogous to ζ_{-1} of UMAP], which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

In most cases $\zeta_1^{(-1)}$ is finite and strictly positive (Fig. 13(b,c)), which places TriMap in the same category as UMAP: it too needs learning-rate annealing or an equivalent mechanism (the TriMap implementation uses delta-bar-delta (Jacobs, 1988) under appropriate initialization).

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends on ζ_1 , ζ_2 , $\cos \theta$, and λ ; here the quantity of interest is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) trace the repulsion behavior as the other quantities vary, with values above (below) the dotted line indicating repulsion (attraction). The striking difference from UMAP is that TriMap’s repulsion can produce contraction rather than expansion. Because this anomaly is confined to small distances, appropriate initialization and triplet selection avoid it.

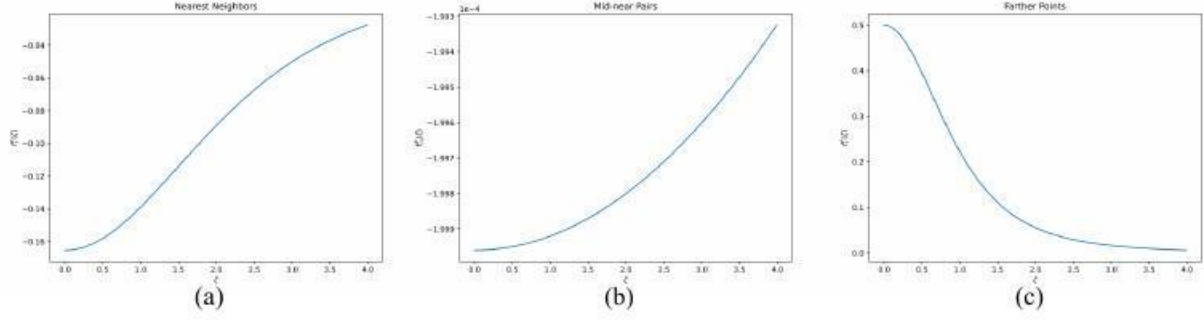


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points (note that the values are on the order of 10^{-4}). (c) Repulsion shape for farther pairs. $\lambda = 1$ for all figures.

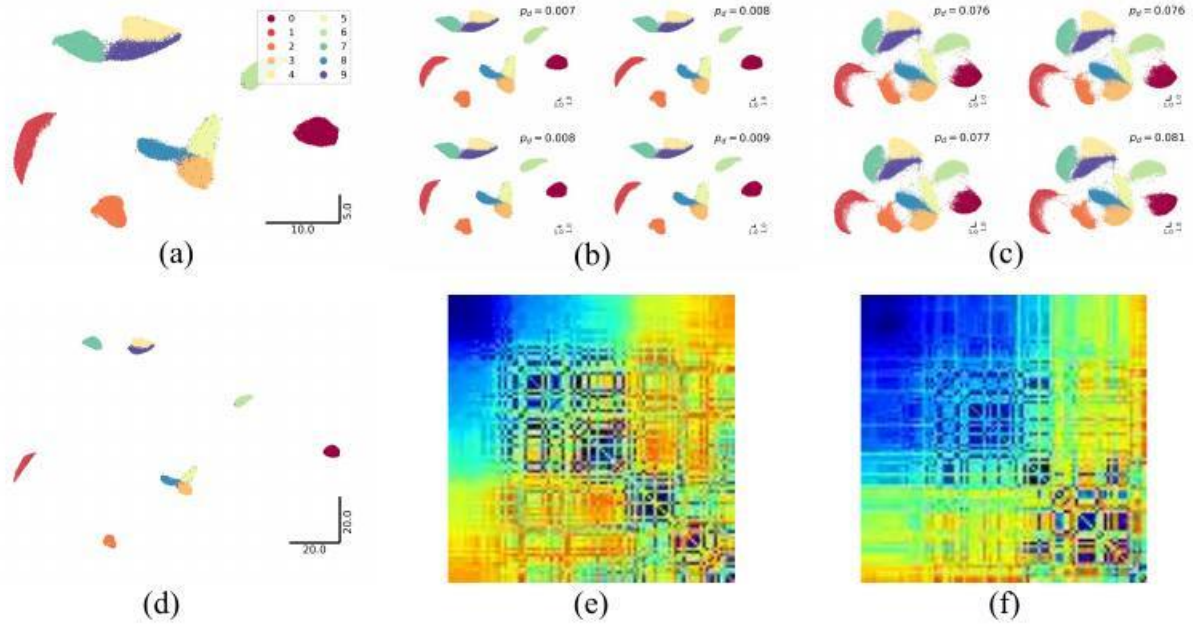


Figure 15: PaCMAP behavior across conditions, showing the same attraction-driven consistency gain seen in UMAP. (a) PaCMAP of MNIST with PCA initialization. (b,c) The four randomly initialized runs best matching (a) for (b) original PaCMAP and (c) PaCMAP with a modified attraction shape, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP of MNIST with a modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (mean $p_d = 0.51 \pm 0.22$) and (f) PaCMAP with modified attraction (0.43 ± 0.22 , a 16% reduction). As in UMAP, strengthening attraction at farther distances improves run-to-run consistency, while increasing repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes low-dimensional embedding at different scales. The loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} are weights of the nearest neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms provide attraction, whereas the last term provides repulsion. A closer look at the loss function reveals that the function is a modified form of TriMap’s triplet loss. For the attractive terms, it replaces TriMap’s affinity for distant points with constant terms (1/10 for nearest neighbors, 1/10000 for mid-near pairs, and 1 for farther points). This loss function is thus separable into three terms and decouples

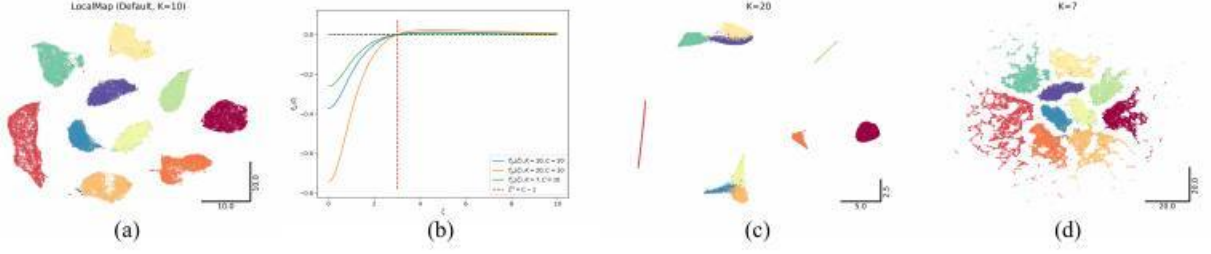


Figure 16: LocalMAP on MNIST, with K tuning the balance between the two forces. (a) Default embedding. (b) Nearest-neighbor attraction-repulsion shape: values below (above) the dotted line indicate attraction (repulsion), with the crossover at $\zeta = \sqrt{C-1}$; following the LocalMAP implementation we set $C = 10$. (c) At large $K = 20$, repulsion dominates and clusters become compact. (d) At small $K = 7$, attraction dominates and clusters break apart. The two panels bracket the default behavior and show that a single parameter shifts the embedding monotonically from compaction to fragmentation.

the update equations. The update equations of the nearest neighbor term are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P follow Proposition 4.1 and f_r follows Proposition 4.2 (Fig. 14). PaCMAP’s attraction is much weaker than UMAP’s, yet this suffices across a wide range of learning rates (modulated by Adam (Kingma and Ba, 2015)); with $w_{NB} = 3$ its maximum attraction stays below 0.5, ruling out flips during the attractive update—the same property that made PaCMAP the strongest local-structure method in Fig. 1. PaCMAP typically initializes within a small low-dimensional sphere (on the order of 10^{-3} in (Wang et al., 2021)) and leans on repulsion to separate clusters. The comparison to UMAP is favorable on both axes: PaCMAP recovers UMAP’s clustering (especially on MNIST) with improved ordering, and its run-to-run consistency already exceeds UMAP’s (Fig. 15(b,e)); adding modified attraction improves it further, from a mean p_d of 0.51 to 0.43 (Fig. 15(c,f)). Adding a small constant to the repulsion shape makes the embedding more compact by widening inter-cluster distance—exactly the behavior observed for UMAP.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) is an iteration of the PaCMAP algorithm. Its defining empirical feature is the clean separation of all 10 MNIST clusters (Fig. 16(a)), matching the behavior we induced by hand in Figs. 3(g,h,j,k) and Figs. 4(a,b). We use this as a further test of our framework, tracing how the attraction-repulsion interplay controls cluster compactness and connectedness.

LocalMAP performs PaCMAP and then does additional optimization on the attraction to decouple some clusters. To this end, it minimizes the following loss function

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term amalgamates the attractive and repulsive nature of the triplet loss function that works on the same pair. In one regime, this function causes attraction, while in the other, it causes repulsion. The second term is identical to the ones in PaCMAP; the only difference is that the algorithm resamples further pairs every few iterations. Thus, we analyze only the first term involving nearest neighbors. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)}. \quad (69)$$

The update provides contraction as long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). When $\zeta^2 > C-1$, $f_{ar} > 0$, and by Proposition 4.2 the update equations provide expansion. The values of K and C determine whether attraction or repulsion dominates the dynamics. In LocalMAP implementation, both C and K are set to 10 (Fig. 16(b)), and the strength of attraction is higher than PaCMAP (Fig. 14(a)). As a result, when $\zeta > 3$, the nearest neighbors face repulsion, causing pairs bridging two clusters to separate. The value of K , working as a scaling parameter for the forces, regulates this separation.

Using Proposition 4.1 $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the maximum values of K , and the maximum attraction possible by f_{ar}^L , without flipping the placements of the pairs. (We would want to avoid flipping the pairs at the LocalMAP optimization to preserve the ordering from PaCMAP; otherwise, it may inhibit cluster separation.) This simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ gives $K \lesssim 13.44$. The prediction is borne out at both ends of the range: above $K \gtrsim 13.44$ repulsion dominates and clusters compact, but LocalMAP's objective fails (the inter-cluster bridge persists in Fig. 16(c) at $K = 20$), while below the threshold attraction dominates (repulsion is too weak) and existing clusters fragment (Fig. 16(d) at $K = 7$, closely reproducing Fig. 4(b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise distances. The loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair. The original implementation of t -SNE considers all the pairs (not just nearest neighbors). This loss function decomposes into attraction and repulsion forces by

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As previously, the first term provides the attractive forces and the second term provides the repulsive forces. While the attractive term is identical to that of UMAP (with $a = 1$ and $b = 1$) and is easy to compute, the repulsive term is coupled among every pair and thus, is very costly. Using the same principles we applied for UMAP, we can write the update equations of t -SNE. Since the original t -SNE didn't rely on the nearest neighbor graph, the weight $w_{i,j}$, computed for all the pairs, is important in the update equations. The attractive update equations are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive update Eqs. (72)-(73), f_a^{t-SNE} follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and gives the minimum distance for contraction, ζ_{-1}). The repulsive update is coupled with all the pairs and thus does not have a simple relation to the repulsion shape. Rather, enabled by our experience from TriMap’s analysis, we can derive the following:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Since t -SNE’s condition for repulsion (Eq. 78) resembles that of TriMap, the repulsion behavior will be the same. Thus, t -SNE’s repulsive forces can give both attraction and repulsion.

Because t -SNE’s forces are scaled (by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion), both are typically much weaker than UMAP’s. This directly explains why t -SNE requires far larger learning rates—on the order of 10^3 in the Open- t -SNE package (Poličar et al., 2024), versus $\mathcal{O}(1)$ for UMAP. It also explains why most t -SNE implementations require an ‘early exaggeration’ step, multiplying the attractive forces by a constant for the first few iterations. This causes points that are supposed to be closer but currently placed far apart to approach each other (inducing far-sightedness). On the other hand, if some points are very close ($\zeta_{i,j} \approx \zeta_{-1}$) but require separation, this early exaggeration trick achieves that as well. Thus, ‘early exaggeration’ plays a vital role in finding a consistent embedding in t -SNE and is an indispensable feature of the algorithm; especially when initialized randomly. (This trick, when applied throughout the optimization, makes t -SNE embeddings look closer to UMAP embeddings (Böhm et al., 2022).)

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is one of the oldest algorithms in this class. This follows the same formula of t -SNE (Eq. 71), but with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This results in the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

that follows Proposition G.3 (by replacing the t -SNE symbols with SNE counterparts). Here the constant attraction shape $f_a^{SNE} = -2$ is ill-posed: contraction depends entirely on the learning rate λ and weights $w_{i,j}$, so clusters overlap even for few samples—the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and its successors fix this quantitatively by swapping the Gaussian kernel for a heavy-tailed one, i.e., Eq. (43).

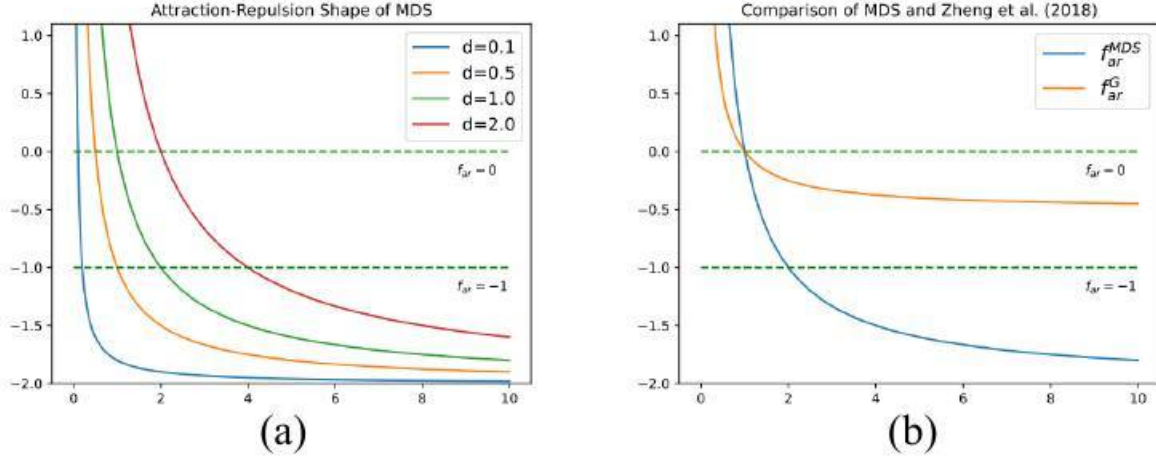


Figure 17: (a) Attraction-repulsion characteristic of MDS via f_{ar} . (b) Side-by-side comparison of the MDS characteristic against the method of (Zheng et al., 2018) for $d = 1.0$, highlighting that the latter’s shape stays within the contracting range $[-0.5, 0]$.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) typically does not use gradient methods, as they often fail to converge to good mappings; instead, it employs stress majorization. Nevertheless, few works discuss gradient methods (Kruskal, 1964; Zheng et al., 2018). We offer a brief treatment for this below. Particularly, (Zheng et al., 2018) formulates a successful gradient descent-based MDS algorithm for graph drawing. We start with the cost function (we can ignore the weight $w_{i,j}$ without loss of generality to their approach, and for discussion, we can lump it into the learning rate):

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss has a singleton term for each pair and does not have explicit attractive and repulsive terms. The term itself will provide both attraction and repulsion. Fortunately, the loss is separable into individual (i, j) components, allowing us to analyze a single pair. The derivative is given by

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. Here, the shape is attractive when $\zeta > d$. In this range, the shape is bounded within $[-2, 0]$ (Fig. 17(a)). However, this is not suitable for convergence as values only within $[-1, 0]$ contract, while the others expand. To make things worse, values lower than -2 cause the points to flip and expand. To work around this, (Zheng et al., 2018) uses an ad-hoc formulation inspired by the force-directed graphs, given by

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which works for an effective learning rate $\lambda \leq 1$: because f_{ar}^G is bounded within $[-0.5, 0]$, any $\lambda \leq 1$ contracts. For $\zeta < d$, $f_{ar}^{MDS}(> 0)$ is instead repulsive. The key contrast with UMAP is quantitative: this attraction-repulsion interaction works well only when the initialization already sits near the target output. Continued optimization of a pair makes it oscillate around $\zeta = d$ (where $f_{ar}^G = 0$), an equilibrium analogous to UMAP’s ζ_{-1} —but here no learning rate drives that distance to zero (indeed, converging to $\zeta = d$ is exactly the MDS objective). This is precisely why MDS clusters tend to be fuzzier than UMAP’s or PaCMAP’s (for relevant illustrations, see (Lambert et al., 2022), (de Bodt et al., 2025), and (Kury et al., 2025)). Note that (Lambert et al., 2022) converts the MDS loss function to a quartet stress (loss involving four samples) and a relative distance (distance divided by the sum of six distances in the quartet), enabling global regularization and reduced computation than the majorization approach.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis \(2002\)](#) underpins modern dimensionality reduction algorithms. It constructs a high-dimensional graph of the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ by the following system of equations:

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match a user-defined value *perplexity*, P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

On the other hand, UMAP constructs its high-dimensional graph by the following system of equations relying on the k-nearest neighbor (k-NN) algorithm:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k -nearest neighbors of the point x_i and σ_i is a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k-NN graph for the affinities (all equal to 1) with a self-tuning distance measure [Zelnik-Manor and Perona \(2004\)](#),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance between x_i and its Euclidean nearest fourth to sixth neighbors. The purpose of this is the same as the corresponding σ_i parameters in Eq. [\(87\)](#) and [\(90\)](#), despite all three being defined differently.

Regardless of these high-dimensional graph choices, our analysis shows that the low-dimensional optimization—specifically the attraction and repulsion shapes—is the primary driver of the resulting embedding, a conclusion that the controlled comparisons throughout this paper make quantitatively concrete.

I Detailed Results for Varying λ_a and λ_b

This section expands the trustworthiness and silhouette comparison of Fig. 1(h,i) into a full, per-method, per-dataset sweep. For a controlled comparison, we hold UMAP’s optimizer fixed and swap in each competing method’s attraction and repulsion shapes, so any metric difference reflects the shapes alone. Figure 18 reproduces the main-text MNIST scores. Figures 19-22 then show, for every method, the embedding at each (λ_a, λ_r) setting, alongside its initialization, an annealed- λ reference, and the two ablations $\lambda_a = 0$ and $\lambda_r = 0$. For PaCMAP we additionally include a mid-near-pairs reference. Figures 23-27 and Figs. 28-32 repeat the entire sweep on FMNIST and single-cell transcriptomes, confirming that the same pattern holds across all three datasets.

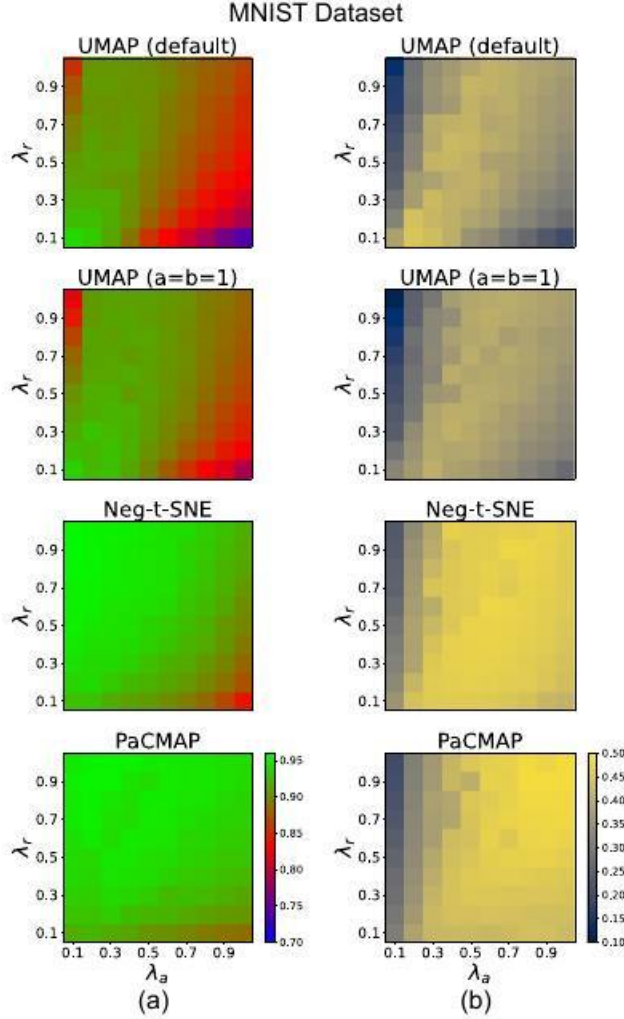


Figure 18: Head-to-head (a) trustworthiness and (b) silhouette score across methods on MNIST, as λ_a and λ_r vary. UMAP peaks on both metrics only at small λ_a , whereas NEG- t -SNE and PaCMAP hold up across the full range.

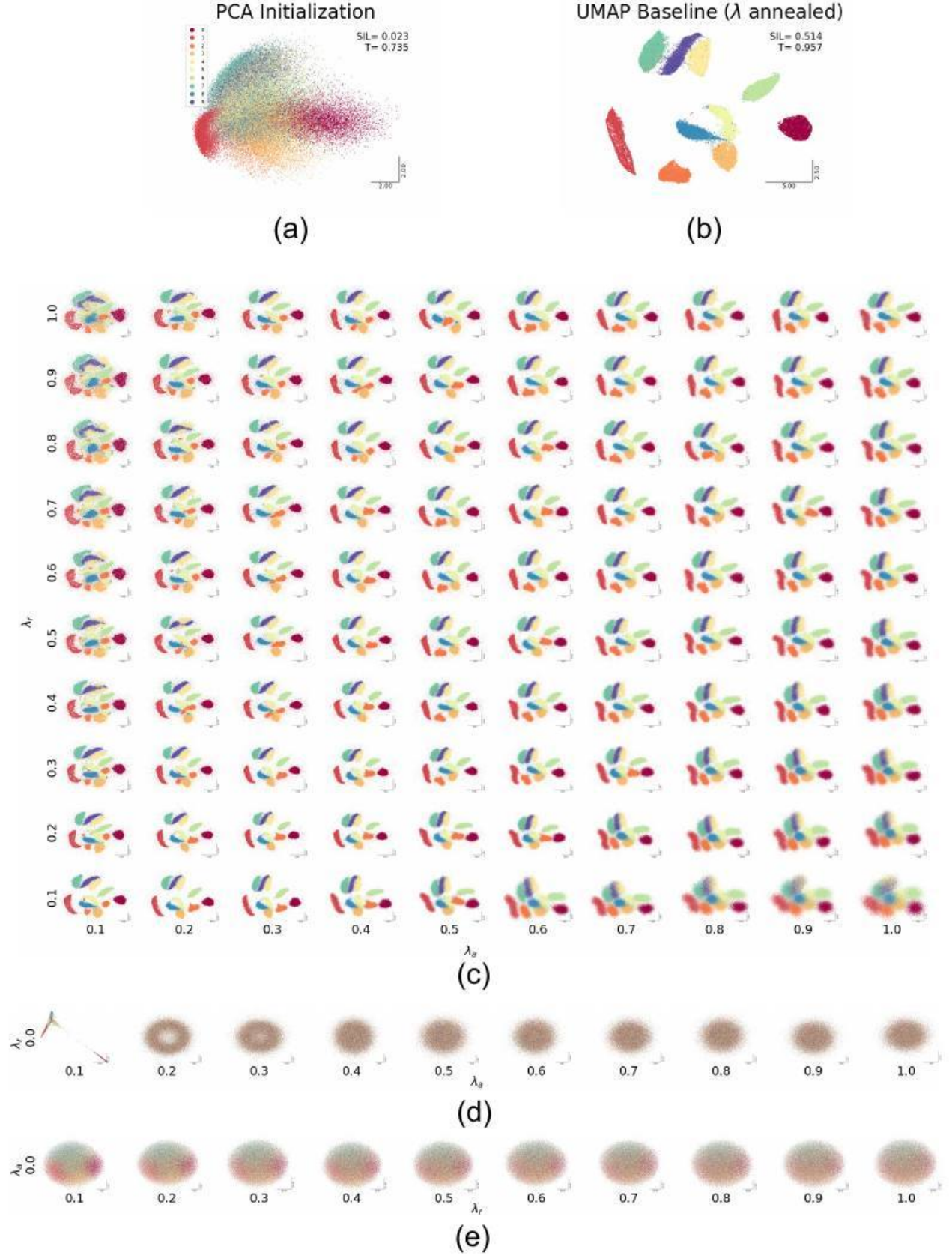


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

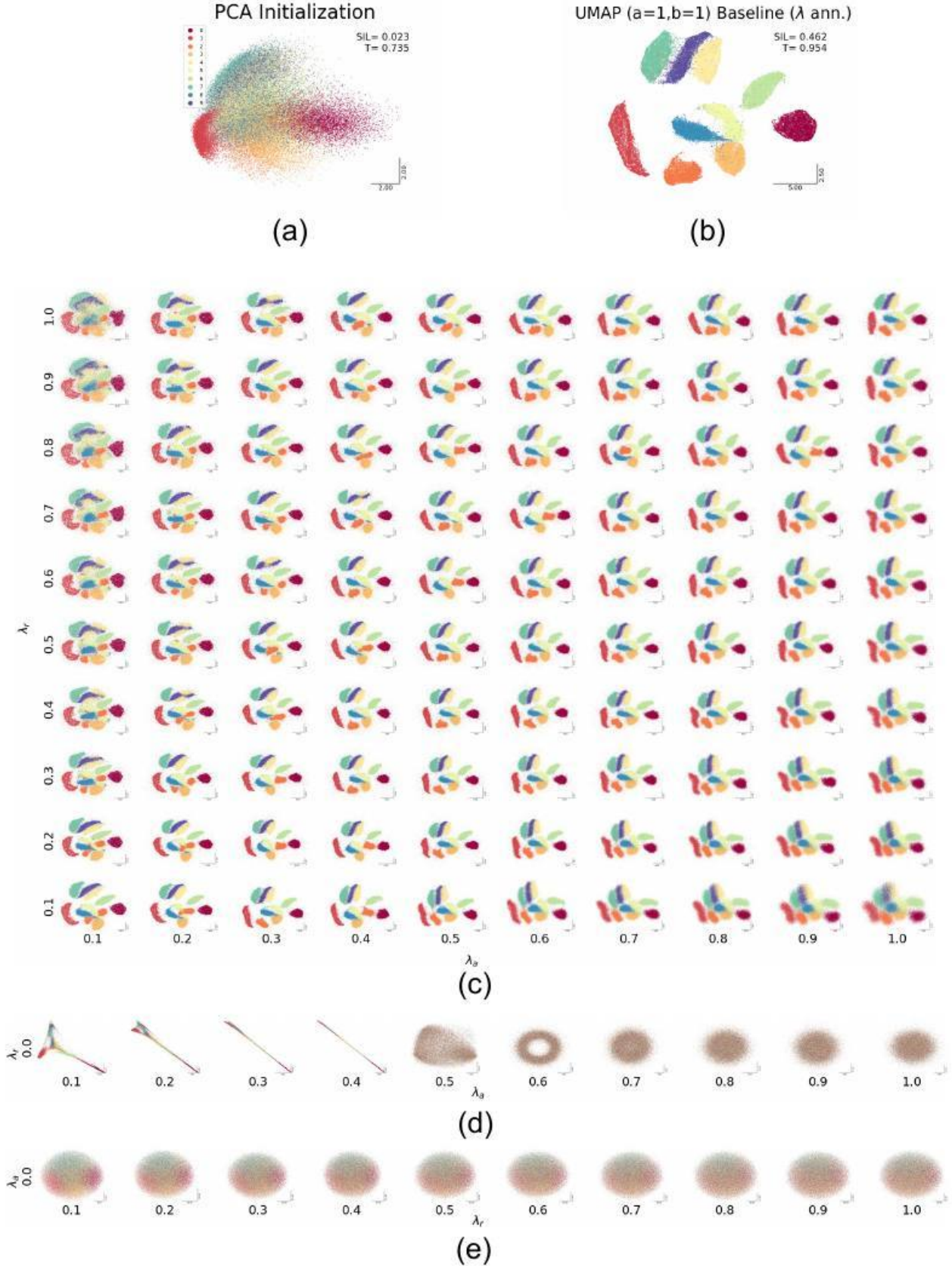


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

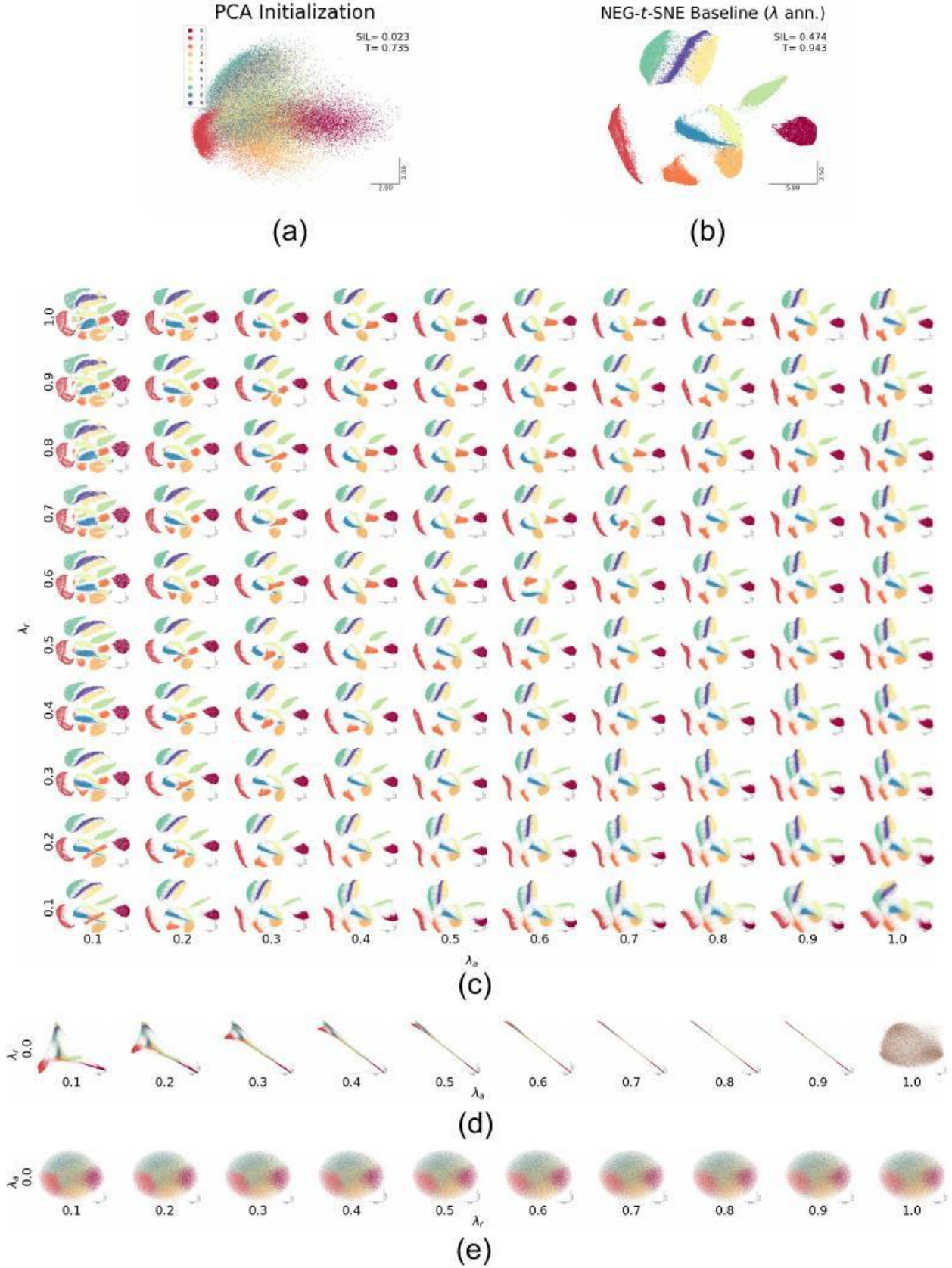


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

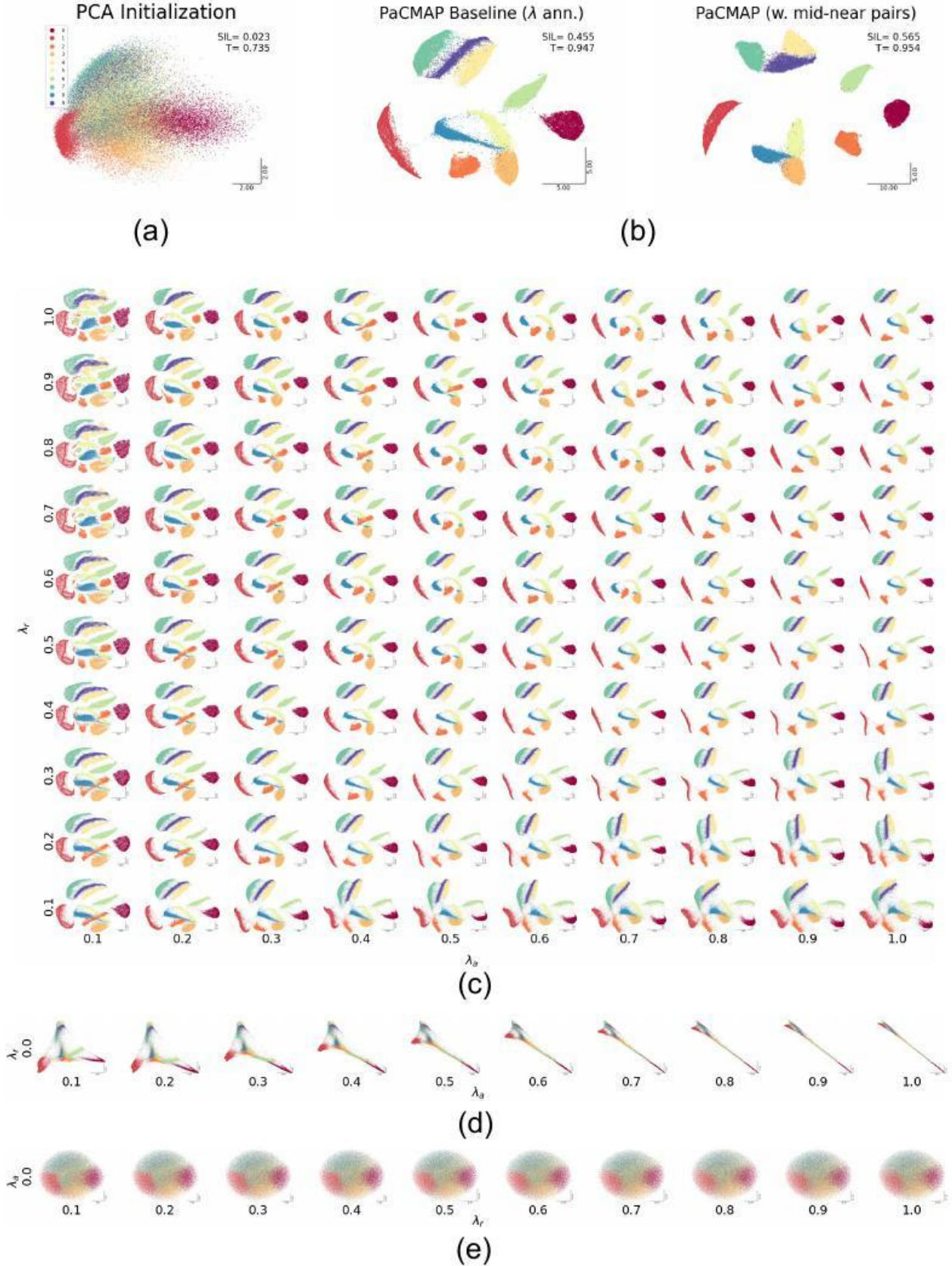


Figure 22: Varying λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

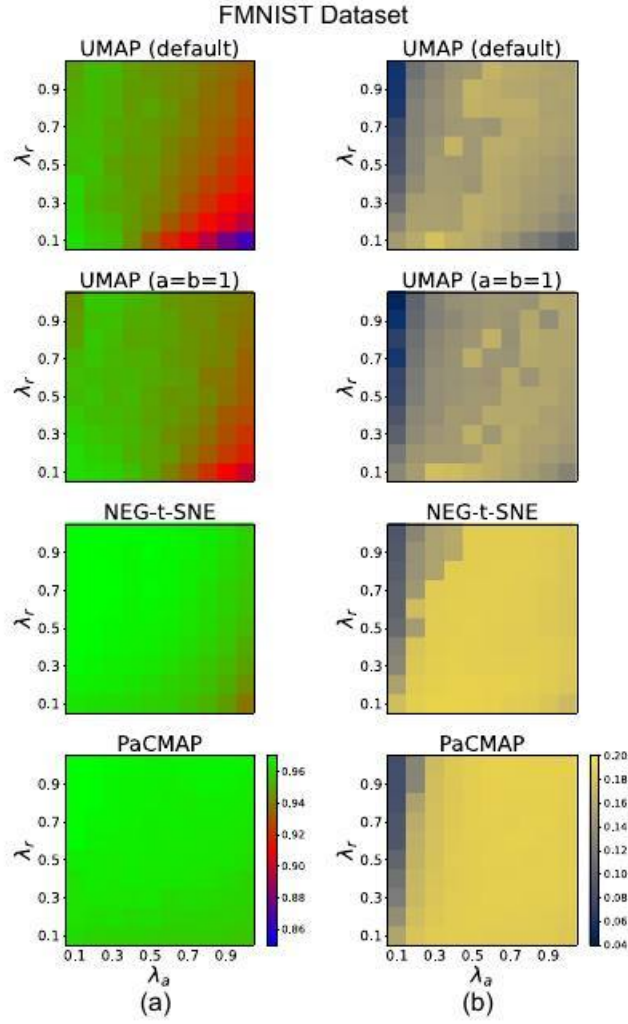


Figure 23: Head-to-head (a) trustworthiness and (b) silhouette score across methods on FMNIST, as λ_a and λ_r vary, reproducing the MNIST pattern: UMAP is sensitive to λ_a while NEG-t-SNE and PaCMAP are not.

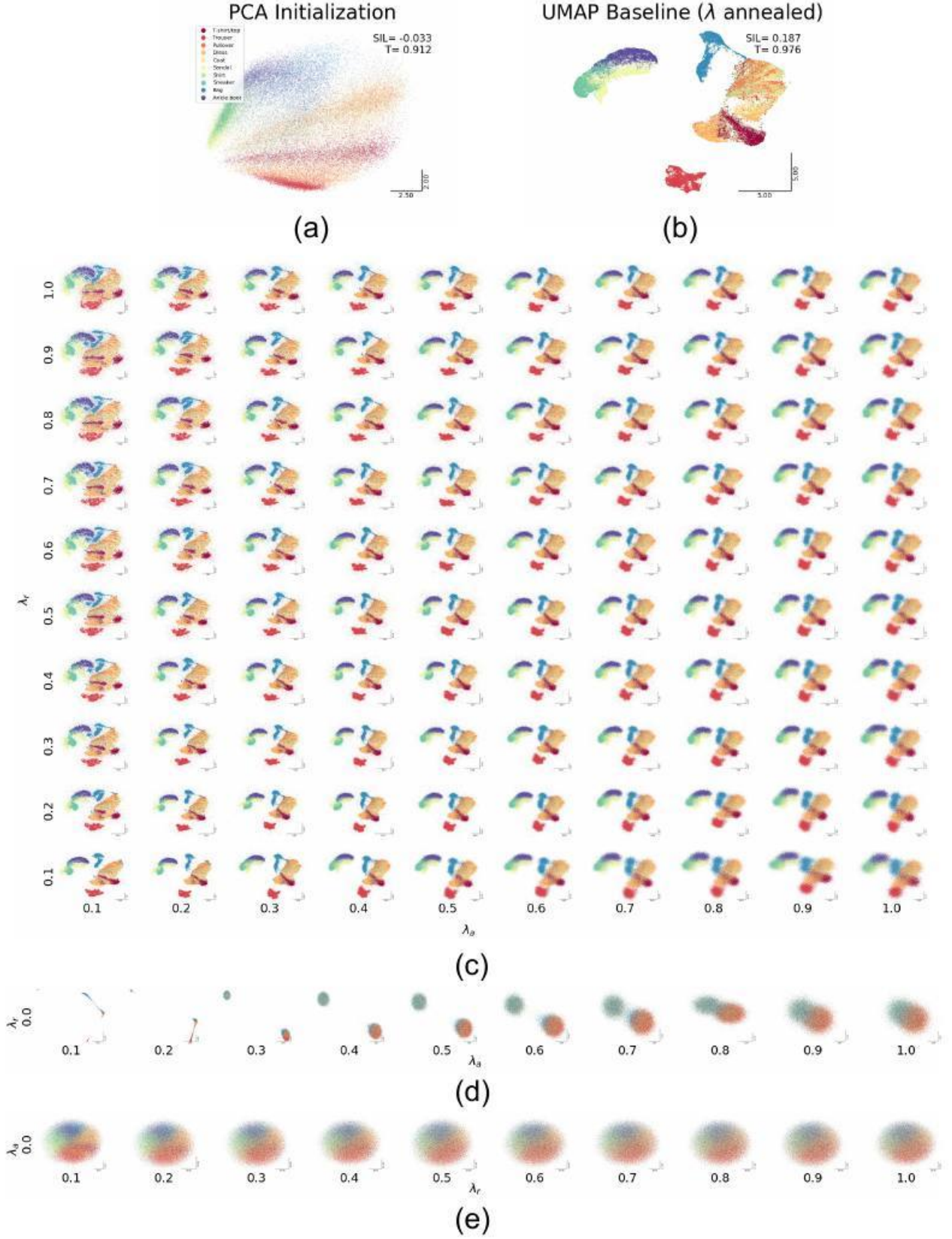


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

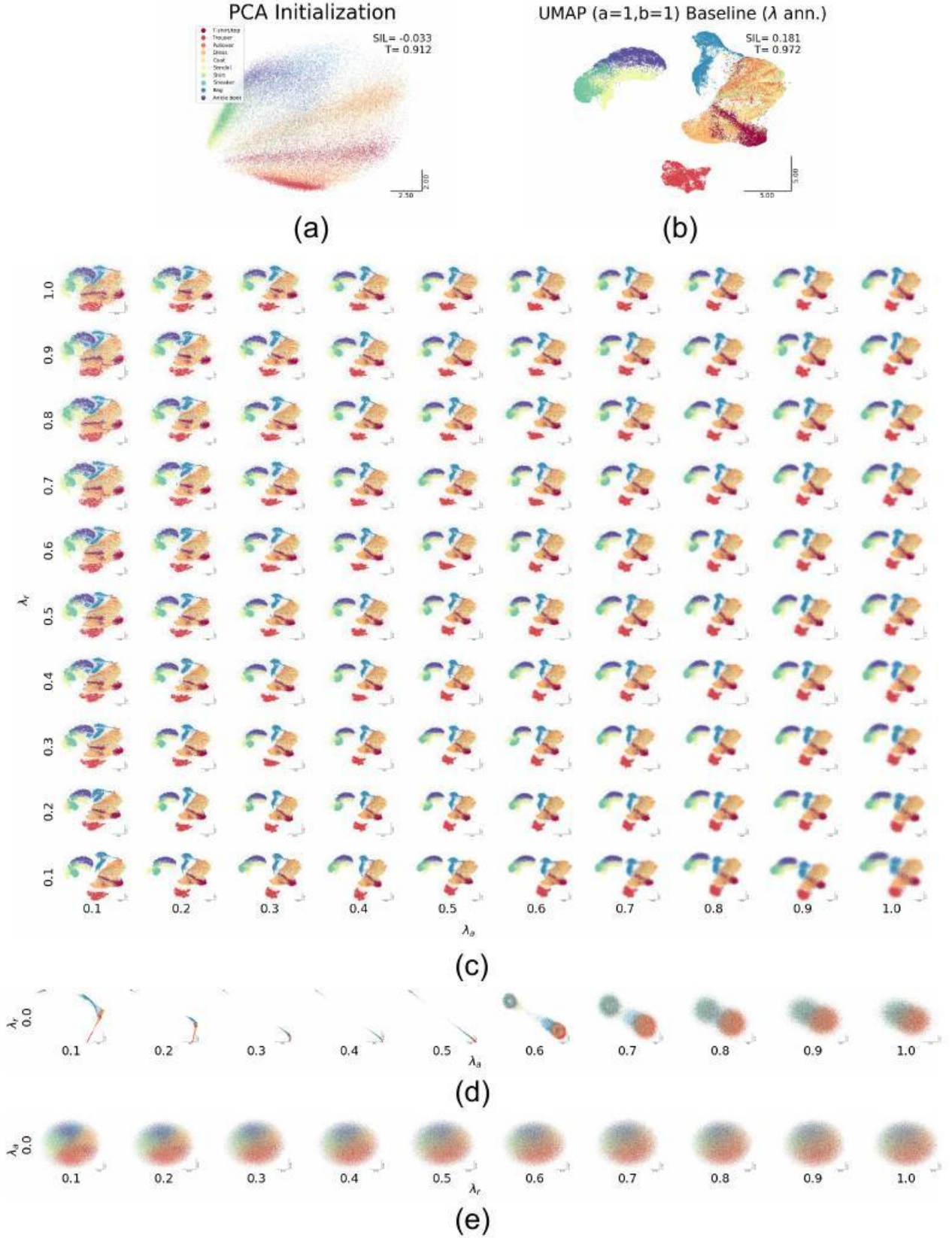


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

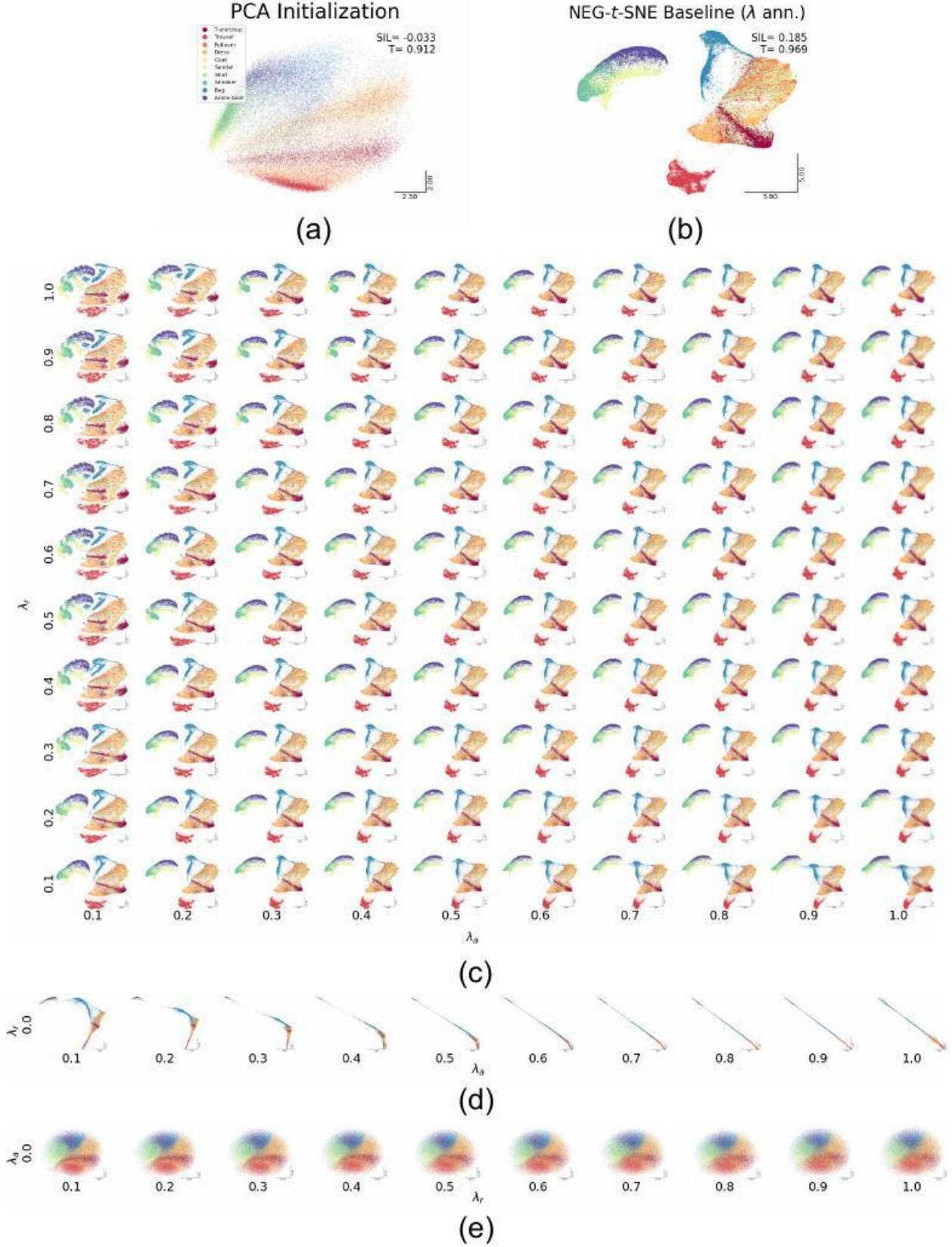


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

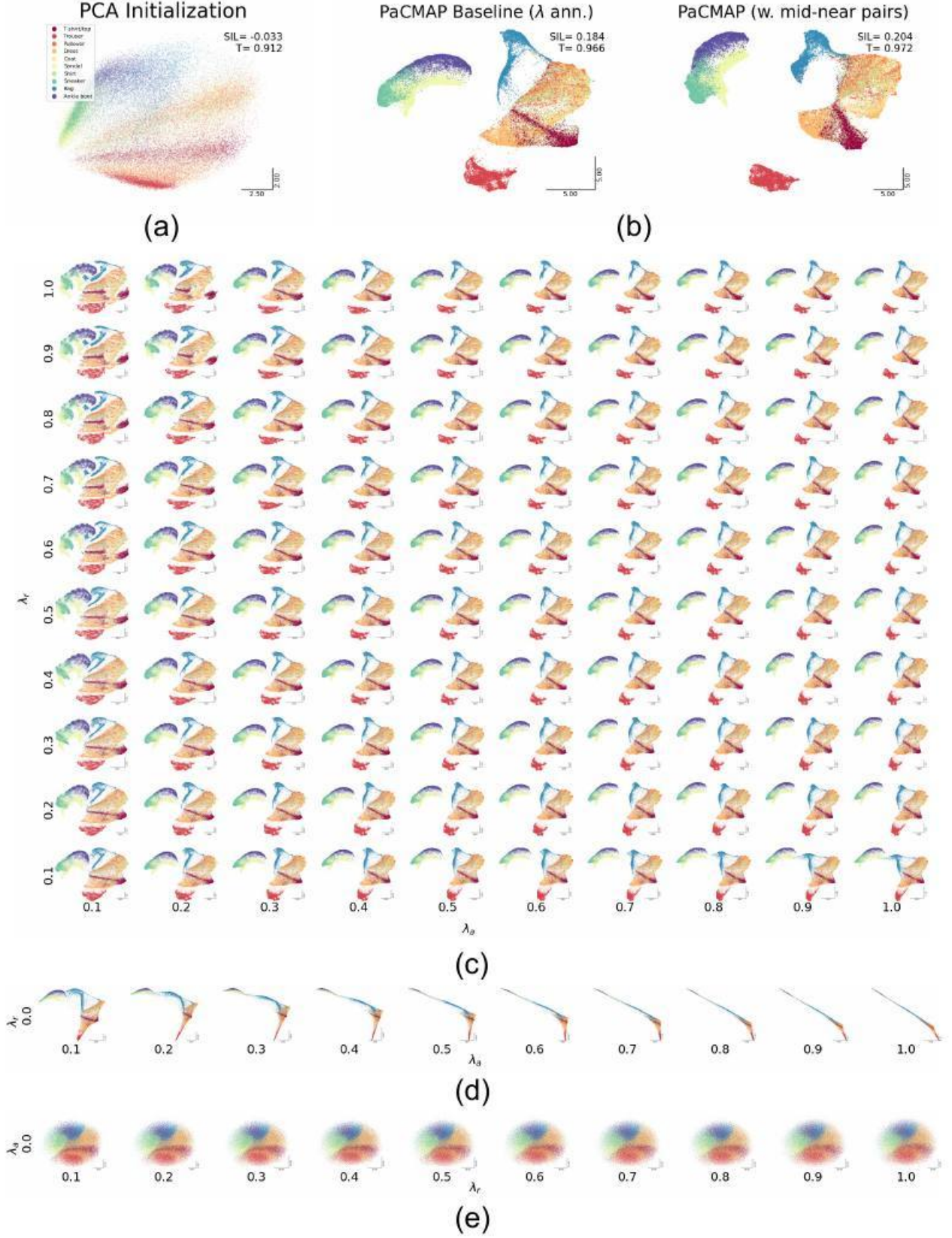


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

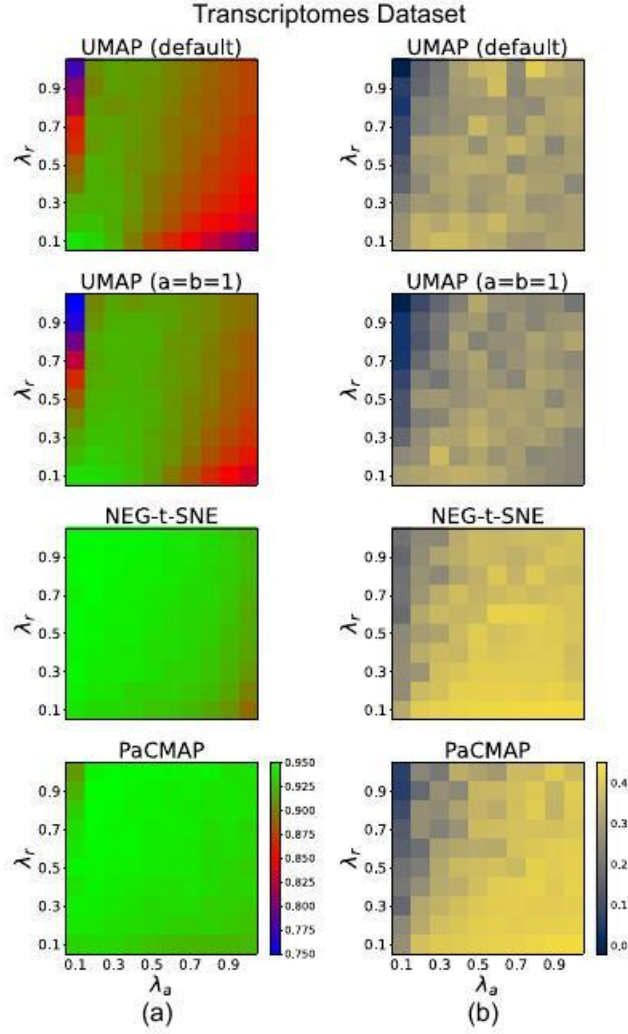


Figure 28: Head-to-head (a) trustworthiness and (b) silhouette score across methods on the single-cell transcriptomes dataset, as λ_a and λ_r vary—the same UMAP-versus-others contrast persists on this third, biologically distinct dataset.

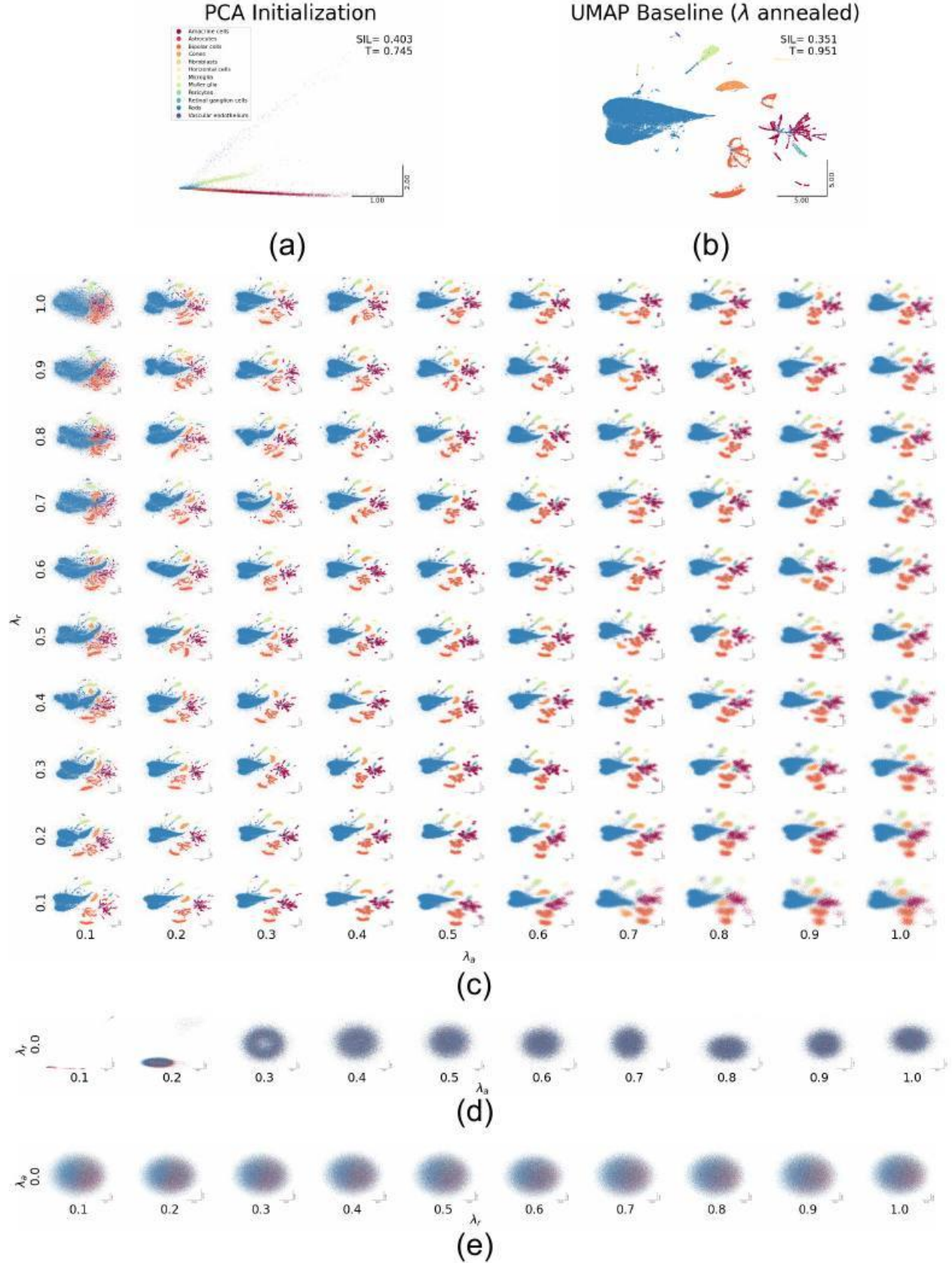


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

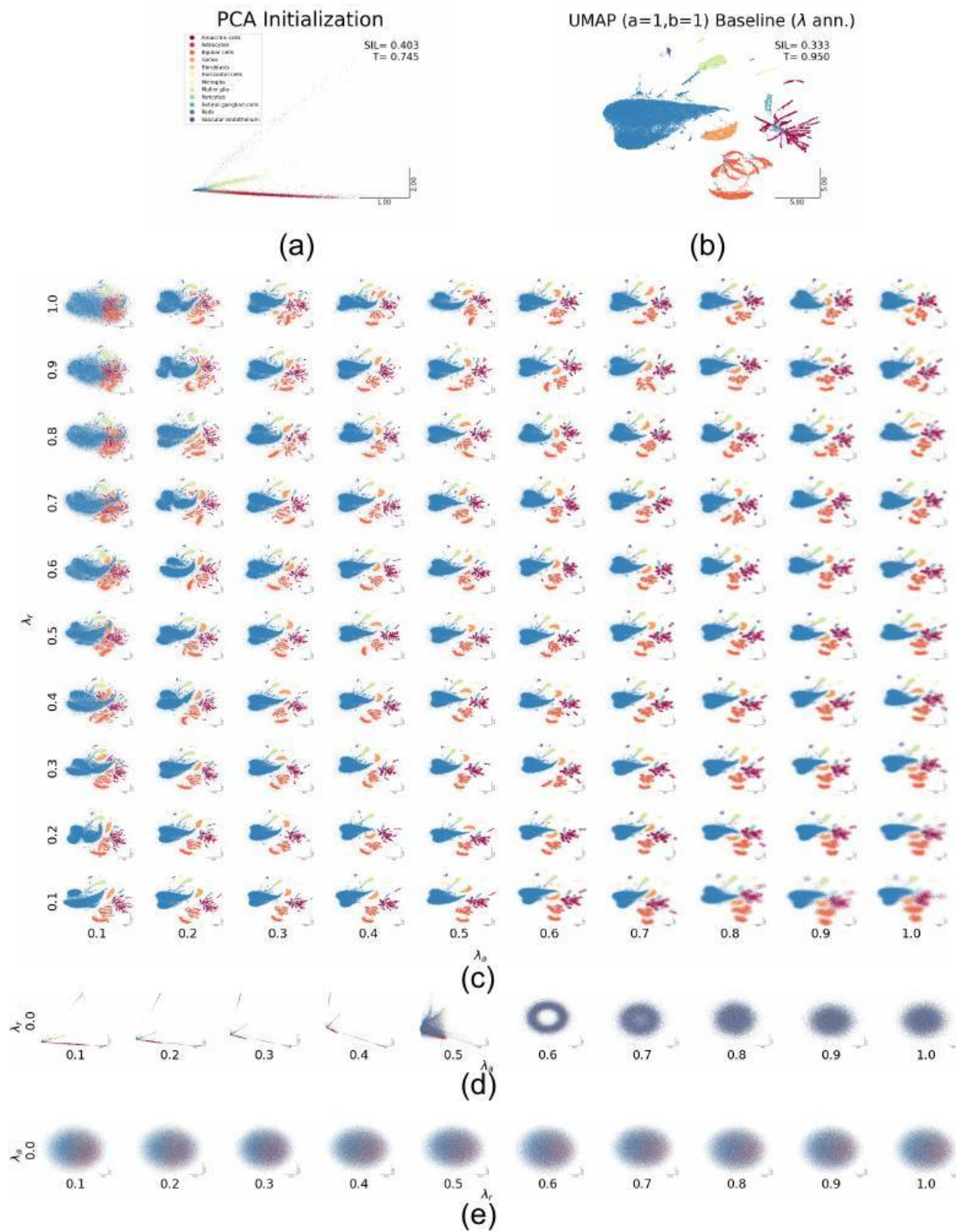


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

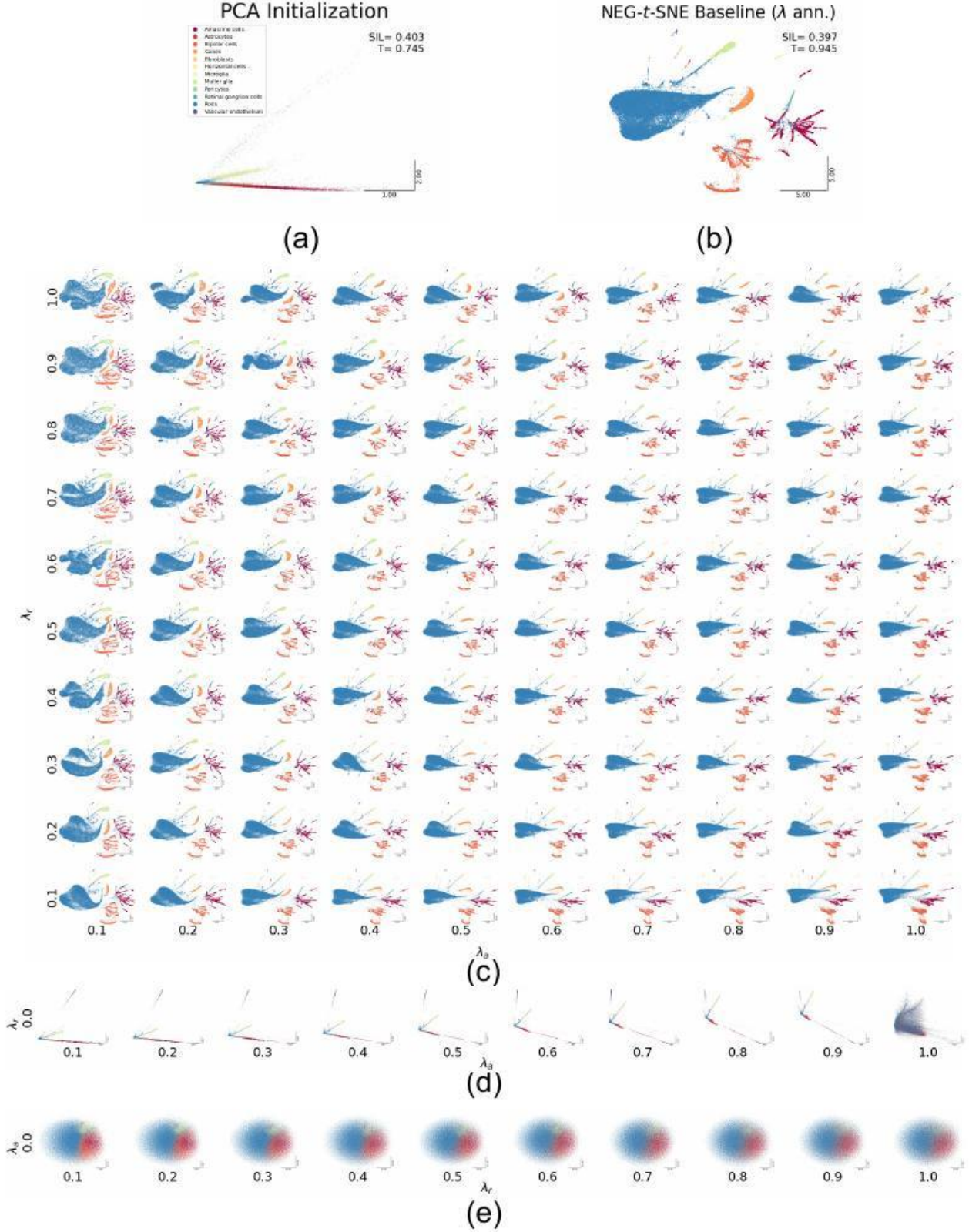


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no distinct clusters, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary.

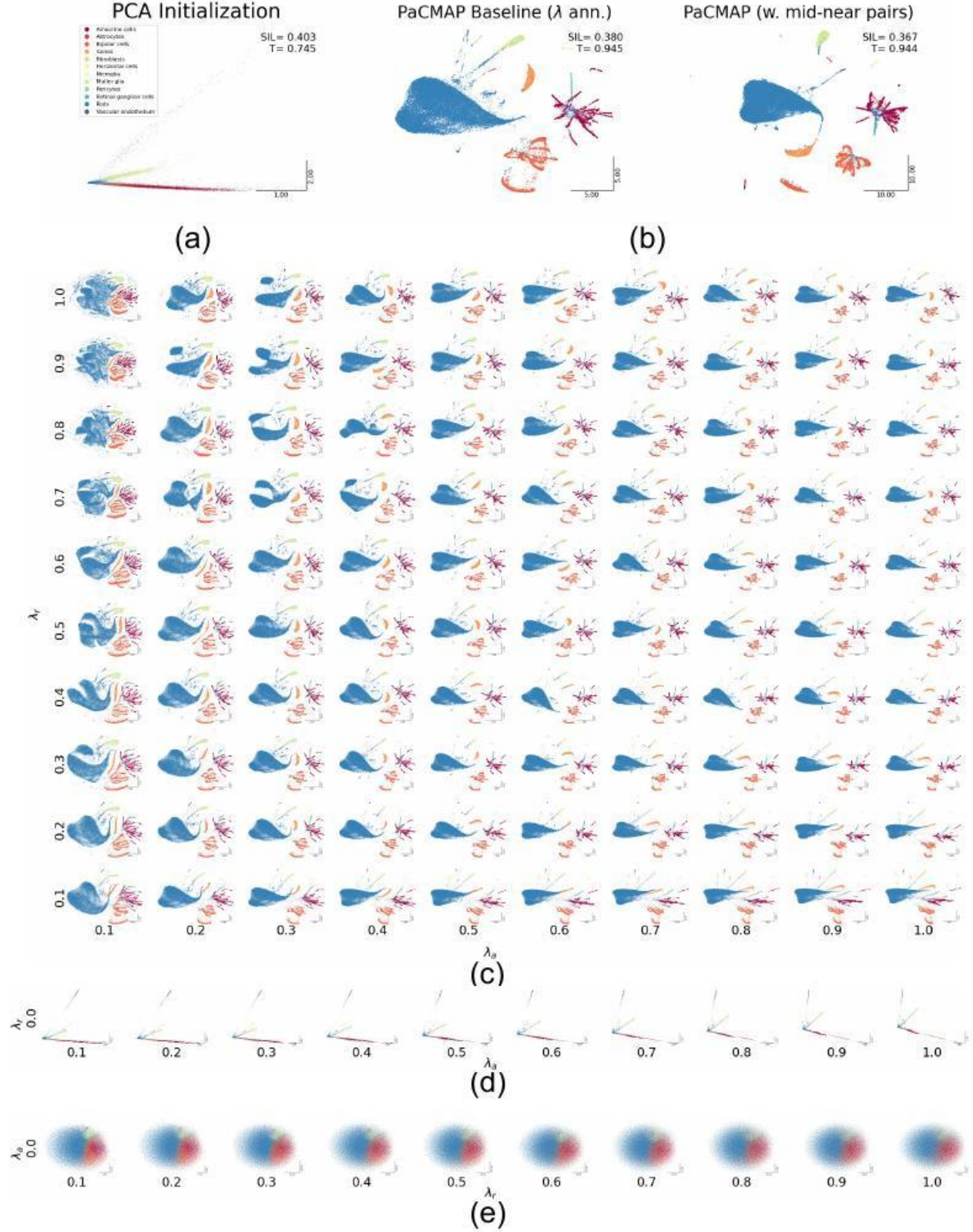


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings as λ_a and λ_r vary (without any annealing). The two single-force ablations are decisive: (d) with $\lambda_r = 0$ (no repulsion), attraction alone produces no cluster, and (e) with $\lambda_a = 0$ (no attraction), repulsion alone likewise produces none—confirming that both forces are jointly necessary. 44

J Mix and Match Approach

Because our analysis treats the attraction and repulsion shapes as independent, they can be freely recombined across loss functions and principles. Figure 33 shows several such combinations. To make the comparison clean, we restricted every attraction shape to $[-1, 0]$ (satisfying Proposition 4.1 hence low learning-rate sensitivity), used only finite repulsion shapes (removing the large short-distance repulsion), held the learning rate constant at $\lambda = 1$, and ran the identical UMAP optimizer for all combinations—so any metric difference is attributable to the shapes alone. The two metrics separate the combinations cleanly: UMAP’s attraction shape (via $f_a = 0.5f_a^U$) gives the best clustering (highest silhouette score), whereas PaCMAP’s attraction shape gives the best local structure (highest trustworthiness), the latter because PaCMAP’s shape stays within $[-0.5, 0]$ and thus never flips a pair. Pushing this further, confining UMAP’s and NEG-*t*-SNE’s attraction shapes to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$) trades clustering performance for improved local structure (Fig. 34)—a controllable, quantifiable knob between the two objectives.

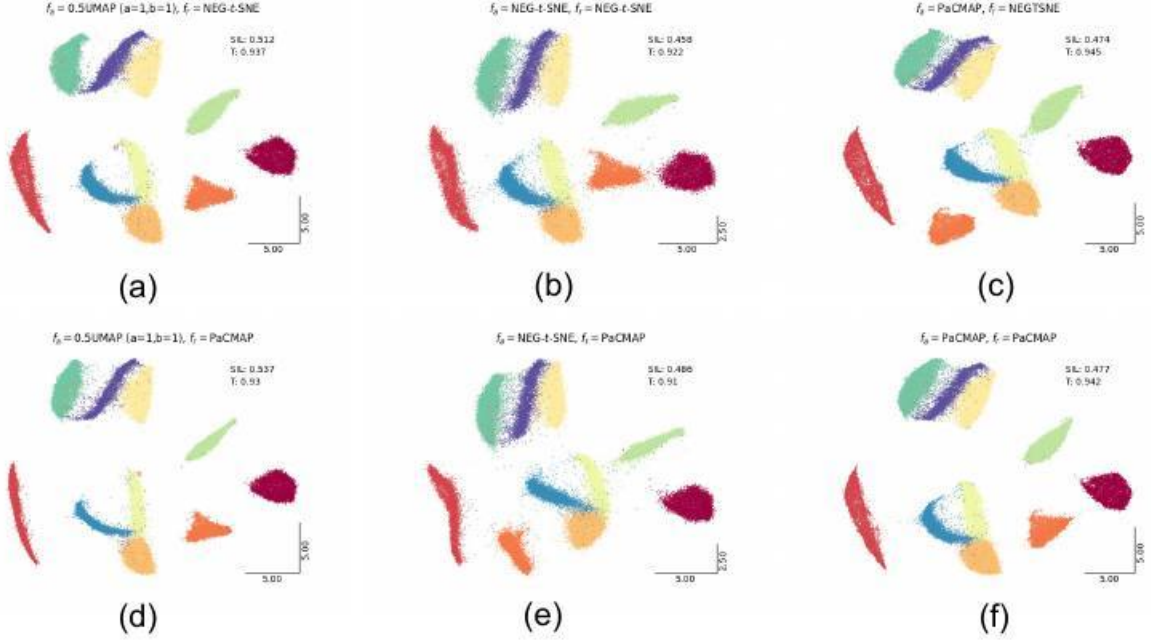


Figure 33: Mixing and matching attraction and repulsion shapes on MNIST at a constant learning rate ($\lambda = 1$), all under the identical UMAP optimizer so differences reflect the shapes alone. Attraction shapes follow Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes, each paired with NEG- t -SNE’s repulsion shape. Bottom row: the same three attraction shapes paired with PaCMAP’s repulsion shape.

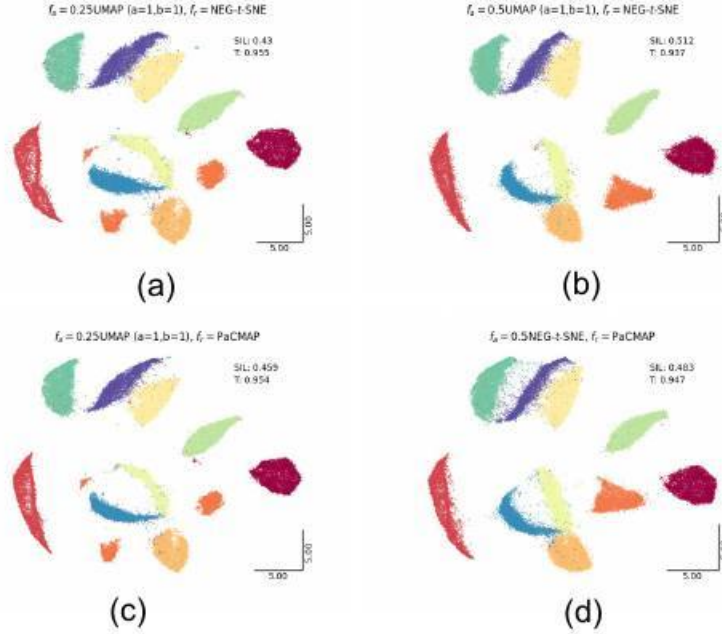


Figure 34: Mixing and matching attraction and repulsion shapes on MNIST at a constant learning rate ($\lambda = 1$), with every attraction shape confined to $[-0.5, 0]$ to rule out flips during the attractive update. Top row: UMAP and NEG- t -SNE attraction shapes, each with NEG- t -SNE’s repulsion shape. Bottom row: the same attraction shapes with PaCMAP’s repulsion shape.

K Implementation Details

For analysis, we implemented our own UMAP algorithm. We used `numba` (Lam et al., 2015) to compute an exact nearest neighbor graph (instead of an approximate one) with $k = 15$ and `scikit-learn`'s (Pedregosa et al., 2011) implementation of the PCA algorithm for PCA initialization. We used this to produce and quantify the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9 and 10. We also used the same implementation when we changed the attraction and the repulsion shapes to those of the alternative methods (Figs. 18, 34, unless otherwise stated). The trustworthiness and silhouette scores were computed using the corresponding function from the `scikit-learn` package.

The mappings shown in Figs. 2, 3, 9, 10 and 15 are rotated to a reference embedding ((a) for each respective Figures). To achieve this, we performed Procrustes alignment of the embeddings by normalizing them (zero mean and unit norm) and then using SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method to extract rotation and scaling parameters.

To compare with Neg- t -SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22(b) (right), 27(b) (right), and 32(b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.